

Proton Conduction at 150°C in a Nanoporous Metal-Organic Framework

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Section S1: Synthetic procedures for the preparation of β -PCMOF2, $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1**, $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**, $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.

General Synthetic Procedures: Phloroglucinol (anhydrous) was obtained from Acros Organics. All other starting materials and solvents were obtained from Aldrich Chemical Co. and used without further purification. Yields reported here have not been optimized.

Synthesis of β -PCMOF2. Anhydrous phloroglucinol (8.0534 g, 63.86 mmol) was dissolved in dry dimethyl carbonate (200 mL) and the resulting yellow solution cooled to 0 °C under Ar. Chlorosulfonic acid (12.43 mL, 186.79 mmol, 2.925 equiv.) was added dropwise over the course of 2 minutes with vigorous stirring, resulting in an orange colored solution. The reaction was warmed to ambient temperature and allowed to proceed for 1.5 h under a steady flow of Ar before the system was sealed and stirred for

an additional 22.5 h. The solvent was then removed via rotary evaporation and the resulting viscous, brown oil dissolved in 150 mL of water. Sodium bicarbonate was added to the solution until it reached a pH of 2.0. Addition of the aqueous solution to 4 L of acetone resulted in the formation of a thick white precipitate, which was isolated via vacuum filtration. Yield: 22.6073 g, 84%. This phase (α) was then converted to the β -phase via heating in a bomb apparatus. As synthesized α -PCMOF2 (0.5000 g, 1.16 mmol) was placed into a 20 mL teflon cup. To this a mixture of water (5 mL), methanol (1.5 mL) and acetone (1.5 mL) was added. The teflon cup was then sealed in a hydrothermal bomb. The bomb was heated at 100 °C for 24 h and then cooled back to room temperature over a period of 12 hours. The resulting white precipitate was isolated via vacuum filtration. Yield: 0.4912 g, 98%.

Synthesis of [β -PCMOF2(Tz)_{0.3}], 1. Prepared following the same procedure as that for β -PCMOF2 except for one additional step, where 1,2,4-triazole was added after preparation of the ligand. Anhydrous phloroglucinol (6.1500 g, 48.77 mmol) was dissolved in dry dimethyl carbonate (175 mL) and the resulting yellow solution cooled to 0 °C under Ar. Chlorosulfonic acid (9.50 mL, 142.64 mmol, 2.925 equiv.) was added dropwise over the course of 2 minutes with vigorous stirring, resulting in an orange colored solution. The reaction was warmed to ambient temperature and allowed to proceed for 1.5 h under a steady flow of Ar before the system was sealed and stirred for an additional 22.5 h. The solvent was then removed via rotary evaporation and the resulting viscous, brown oil dissolved in water to a concentration of 125 mgmL⁻¹. 1,2,4-

Triazole (3.2835 g, 47.54 mmol,) was then added to the aqueous solution and allowed to stir for 24 h. Sodium bicarbonate (5.9921 g, 71.33 mmol, 1.5 equiv.) was added to the solution, which was then precipitated via addition to 3 L of acetone. The resulting thick white precipitate was isolated via vacuum filtration. Yield: 10.6785 g, 61.3%. The α -phase was then quantitatively converted to β -phase in a bomb apparatus. Approximately 2 g of as synthesized α -PCMOF2(Tz)_{0.3} was packed into a silicone tube, sealed and pressed under a hydrostatic pressure of 5000 kg for 2 min. The resulting pellet was removed from the silicone tube and placed into a 20 mL teflon cup, along with a vial containing 0.5 mL of water. The teflon cup was then sealed in a hydrothermal bomb and heated at 80 °C for 8 h. Conversion to the β -phase was confirmed via PXRD analysis.

Synthesis of [β -PCMOF2(Tz)_{0.45}], 2. Prepared following the same procedure as that for β -PCMOF2(Tz)_{0.3} except for the addition of 1.35 equivalents (relative to ligand) of NaHCO₃ during the preparation of the framework. Yield: 10.5333 g, 61.1%. The resulting α -phase was then converted to β -phase via the same procedure as described above for β -PCMOF2(Tz)_{0.3}.

Synthesis of [β -PCMOF2(Tz)_{0.6}], 3. Prepared following the same procedure as that for β -PCMOF2(Tz)_{0.3} except for the addition of 1.20 equivalents (relative to ligand) of NaHCO₃ during the preparation of the framework. Yield: 10.8796 g, 62.7%. The resulting α -phase was then converted to β -phase via the same procedure as described above for β -PCMOF2(Tz)_{0.3}.

Synthesis of deuterated β -PCMOF2(Tz) (d -[β -PCMOF2(Tz)_{0.5}]) for ^2D NMR. First 1,2,4-triazole was deuterated following a procedure modified from McDermott et al.^x 1,2,4-triazole (1.6162 g, 23.40 mmol) was added to a 20 mL teflon cup. To this D₂O (10 g, 499 mmol) was added. The teflon cup was then sealed in a hydrothermal bomb and heated at 200 °C for 4 hours. The deuterated product was isolated as a white powder by rotary evaporation. The percent deuteration was determined to be 95% via NMR. Yield = 1.6662 g, 94%. The d -[β -PCMOF2(Tz)_{0.5}] framework was prepared following a procedure similar to that for β -PCMOF2(Tz)_{0.3} with a few modifications to limit H-D exchange. Anhydrous phloroglucinol (3.0649 g, 24.30 mmol) was dissolved in dry dimethyl carbonate (75 mL) and the resulting yellow solution cooled to 0 °C under Ar. Chlorosulfonic acid (4.73 mL, 71.09 mmol, 2.925 equiv.) was added dropwise over the course of 2 minutes with vigorous stirring, resulting in an orange colored solution. The reaction was warmed to ambient temperature and allowed to proceed for 1.5 h under a steady flow of Ar before the system was sealed and stirred for an additional 22.5 h. The solvent was then removed via rotary evaporation and the resulting viscous, brown oil dissolved in D₂O to a concentration of 175 mgmL⁻¹. Deuterated 1,2,4-triazole (0.3444 g, 4.78 mmol) was then added to 10 mL (1.7500 g, 4.78 mmol) of the solution and allowed to stir for 24 h. Sodium carbonate (0.3038 g, 2.87 mmol, 1.2 equiv.) was added to the solution, which was then precipitated via addition to 300 mL of freshly distilled acetone. The resulting thick white precipitate was isolated via vacuum filtration with a swivel frit. Yield: 0.6903 g, 33.4%. The α -phase was then quantitatively converted to β -phase in a bomb apparatus. Approximately 0.6 g of as synthesized d -[α -PCMOF2(Tz)_{0.5}] was

packed into a silicone tube, sealed and pressed under a hydrostatic pressure of 5000 kg for 2 min. The resulting pellet was removed from the silicone tube and placed into a 20 mL teflon cup, along with a vial containing 0.25 mL of D₂O. The teflon cup was then sealed in a hydrothermal bomb and heated at 80 °C for 8 h. Conversion to the β -phase was confirmed via PXRD analysis.

Section S2: Single Crystal XRD of PCMOFs.

Single-crystal diffraction experiments were conducted at 173K for β -PCMOF2. Crystallographic data have been deposited with the Cambridge Crystallographic Data Centre with the number CCDC 746416. Samples were collected using a Nonius Kappa CCD diffractometer with graphite monochromated Mo K α radiation ($\lambda = 0.71069 \text{ \AA}$) and equipped with a CCD detector. The intensity data collection was performed in the ω - ϕ scanning mode with the goniometer and detector angular settings optimized using the program COLLECT.¹ The crystal-to-detector distance was 30 mm. The unit cell and the orientation matrix were refined using the entire data set of reflections. The diffraction spots were measured in full with a high accuracy, scaled with SCALEPACK,¹ corrected for Lorentz-polarization correction, and integrated using DENZO.¹ The crystallographic data and conditions for structure analysis are listed in Table 1. Structure solutions and refinements were run in SIR97² and SHELXL-97³, respectively. The structure model (non-hydrogen atoms) was readily obtained from successive difference Fourier maps. At the end of refinement, anisotropic displacement parameters and extinction coefficients were allowed to vary, with the resulting residuals listed in Table 1.

1. COLLECT, DENZO, SCALEPACK, SORTAV, Kappa CCD Program Package; Nonius BV: Delft, The Netherlands, 1998.

2. Altomare, A.; Burla, M. C.; Camalli, M.; Cascarano, G. L.; Giacovazzo, C.; Guagliardi, A.; Moliterni, A. G. G.; Polidori, G.; Spagna, R. *J. Appl. Crystallogr.* **1999**, 32, 115.

3. Sheldrick, G. M. *SHELXL-97, Program for crystal structure refinement*: University of Göttingen: Germany, 1997.

Table S1. Crystal data and structure refinement for β -PCMOF2.

Empirical formula	C12 Na6 O24 S6		
Formula weight	858.42		
Temperature	293(2) K		
Wavelength	0.71073 Å		
Crystal system, space group	Trigonal, P3*		
Unit cell dimensions	a = 9.5120(13) Å	alpha = 90 °	
	b = 9.5120(13) Å	beta = 90 °	
	c = 3.9870(8) Å	gamma = 120 °	
Volume	312.41(9) Å ³		
Z, Calculated density	6, 2.281 Mg/m ³		
Absorption coefficient	0.769 mm ⁻¹		

F(000)	213
Crystal size	0.1 x 0.06 x 0.06 mm
Theta range for data collection	4.29 to 27.48 deg.
Limiting indices	-12<=h<=12, -12<=k<=12, -5<=l<=4
Reflections collected / unique	2660 / 946 [R(int) = 0.0611]
Completeness to theta = 27.48	99.8 %
Absorption correction	None
Refinement method	Full-matrix least-squares on F ²
Data / restraints / parameters	946 / 104 / 144
Goodness-of-fit on F ²	1.117
Final R indices [I>2sigma(I)]	R1 = 0.0324, wR2 = 0.0652
R indices (all data)	R1 = 0.0418, wR2 = 0.0689
Absolute structure parameter	0.16(13)
Largest diff. peak and hole	0.229 and -0.223 e.A ⁻³

* Platon indicates higher symmetry may be present. In fact, the structure does solve statistically better in a hexagonal setting (R1 < 3%) but the structure solution requires disorders with bond angles that are not chemically rational .

Figure S1. Thermal Ellipsoid Plot of β -PCMOF2. Ellipsoids of 50% probability are shown.

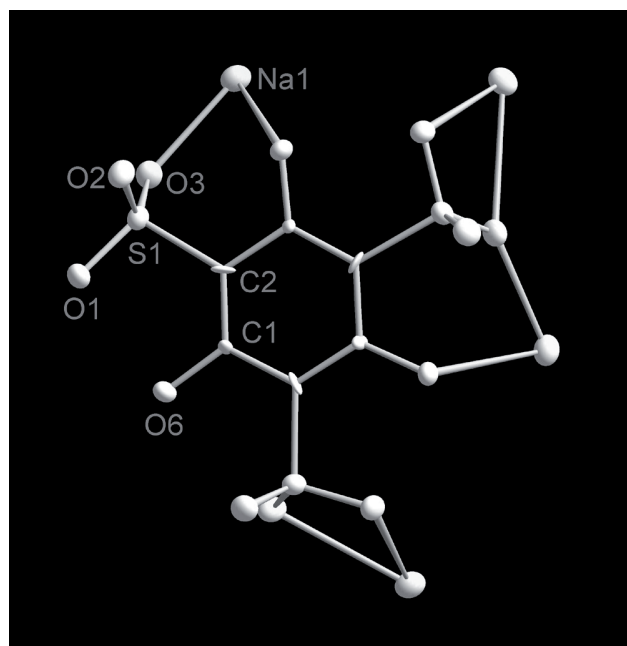


Table S2. Atomic coordinates ($\times 10^4$) and equivalent isotropic displacement parameters ($\text{\AA}^2 \times 10^3$) for β -PCMOF2 in P3. $U(\text{eq})$ is defined as one third of the trace of the orthogonalized U_{ij} tensor.

Moiety	x	y	z	U(eq)
Na(1)	6666(2)	-2172(1)	14442(3)	27(1)
S(1)	7893(2)	777(2)	9450(3)	19(1)
C(1)	5507(16)	1667(11)	9446(19)	20(6)
C(2)	5018(13)	2846(17)	9431(18)	15(5)
O(1)	6420(18)	-839(13)	9410(40)	23(3)
O(2)	8802(5)	1030(5)	12483(9)	25(2)
O(3)	8804(5)	1030(5)	6395(9)	26(2)
O(6)	4324(5)	86(5)	9453(10)	22(2)
S(1')	5444(2)	-444(2)	9445(3)	19(1)
C(1')	7817(16)	2823(17)	9484(19)	10(5)
C(2')	8302(14)	4493(15)	9444(19)	14(5)
O(1')	6905(18)	-621(17)	9470(40)	22(3)
O(2')	4530(5)	-1104(5)	12495(9)	26(2)
O(3')	4526(5)	-1106(5)	6394(9)	25(2)
O(6')	9006(5)	2435(5)	9465(10)	21(2)

Table S3. Bond lengths for β -PCMOF2 in P3.

Moiety	Distance ($\text{\AA}$)
Na(1)-O(3')#1	2.346(4)
Na(1)-O(2')#2	2.348(4)
Na(1)-O(3)#3	2.350(4)
Na(1)-O(2)#4	2.352(4)
Na(1)-O(1')	2.413(14)
Na(1)-O(1)#5	2.425(15)
Na(1)-O(1')#5	2.432(14)
Na(1)-O(1)	2.447(15)
Na(1)-O(6')#4	2.668(4)
Na(1)-O(6)#2	2.669(4)
Na(1)-O(6)#1	2.676(4)
Na(1)-O(6')#3	2.682(4)
S(1)-O(2)	1.435(4)
S(1)-O(3)	1.444(4)
S(1)-O(1)	1.474(12)
S(1)-C(2)#6	1.785(11)
S(1)-Na(1)#7	3.1537(18)
C(1)-O(6)	1.355(10)
C(1)-C(2)#6	1.390(15)
C(1)-C(2)	1.413(12)
C(2)-C(1)#8	1.390(15)
C(2)-S(1)#8	1.785(11)
O(1)-Na(1)#7	2.425(15)
O(2)-Na(1)#9	2.352(4)
O(3)-O(2)#7	1.560(5)
O(3)-O(6')	1.750(5)
O(3)-Na(1)#10	2.350(4)
O(3)-Na(1)#7	2.798(4)
O(6)-Na(1)#11	2.669(4)
O(6)-Na(1)#12	2.676(4)
S(1')-O(2')	1.443(4)
S(1')-O(3')	1.445(4)
S(1')-O(1')	1.481(12)
S(1')-Na(1)#7	3.1537(17)
C(1')-O(6')	1.354(10)
C(1')-C(2')	1.415(12)
O(1')-Na(1)#7	2.432(14)
O(2')-Na(1)#11	2.348(4)
O(3')-Na(1)#12	2.346(4)
O(3')-Na(1)#7	2.800(4)
O(6')-Na(1)#9	2.668(4)
O(6')-Na(1)#10	2.682(4)

Table S4. Bond angles for β -PCMOF2 in P3.

Moiety	Angle ($^{\circ}$)
O(3)#3-Na(1)-O(1)#5	94.4(4)
O(2)#4-Na(1)-O(1)#5	128.2(4)
O(3)#3-Na(1)-O(1)	128.0(4)
O(2)#4-Na(1)-O(1)	94.1(4)
O(1)#5-Na(1)-O(1)	109.8(5)
O(3)#3-Na(1)-O(6)#2	100.16(14)
O(2)#4-Na(1)-O(6)#2	71.45(13)
O(1)#5-Na(1)-O(6)#2	158.6(4)
O(1)-Na(1)-O(6)#2	73.0(3)
O(3)#3-Na(1)-O(6)#1	71.49(13)
O(2)#4-Na(1)-O(6)#1	100.24(14)
O(1)#5-Na(1)-O(6)#1	73.3(3)
O(1)-Na(1)-O(6)#1	158.6(3)
O(6)#2-Na(1)-O(6)#1	96.48(14)
O(2)-S(1)-O(3)	114.9(3)
O(2)-S(1)-O(1)	111.3(6)
O(3)-S(1)-O(1)	110.0(7)
O(2)-S(1)-C(2)#6	107.9(3)
O(3)-S(1)-C(2)#6	107.4(3)
O(1)-S(1)-C(2)#6	104.7(8)
O(6)-C(1)-C(2)#6	123.4(13)
O(6)-C(1)-C(2)	117.4(12)
C(2)#6-C(1)-C(2)	119.2(11)
C(1)#8-C(2)-C(1)	120.8(11)
C(1)#8-C(2)-S(1)#8	122.5(11)
C(1)-C(2)-S(1)#8	116.7(11)
S(1)-O(1)-Na(1)#7	105.4(8)
S(1)-O(1)-Na(1)	104.1(8)
Na(1)#7-O(1)-Na(1)	109.8(5)
S(1)-O(2)-O(3)#5	147.4(3)
S(1)-O(2)-Na(1)#9	130.6(2)
O(3)#5-O(2)-Na(1)#9	70.56(18)
S(1)-O(2)-Na(1)	90.24(18)
O(3)#5-O(2)-Na(1)	73.81(19)
Na(1)#9-O(2)-Na(1)	139.19(16)
S(1)-O(3)-O(2)#7	147.5(3)
S(1)-O(3)-Na(1)#10	130.4(2)
O(2)#7-O(3)-Na(1)#10	70.69(18)
S(1)-O(3)-Na(1)#7	90.25(17)
O(2)#7-O(3)-Na(1)#7	73.81(19)
Na(1)#10-O(3)-Na(1)#7	139.31(16)
C(1)-O(6)-Na(1)#11	121.5(5)
C(1)-O(6)-Na(1)#12	121.2(5)
Na(1)#11-O(6)-Na(1)#12	96.48(14)

Symmetry transformations used to generate equivalent atoms:

#1 -x+y+1,-x,z+1 #2 -x+y+1,-x,z #3 -y+1,x-y-1,z+1
 #4 -y+1,x-y-1,z #5 x,y,z+1 #6 -y+1,x-y,z
 #7 x,y,z-1 #8 -x+y+1,-x+1,z #9 -x+y+2,-x+1,z
 #10 -x+y+2,-x+1,z-1 #11 -y,x-y-1,z
 #12 -y,x-y-1,z-1

Table S5. Anisotropic displacement parameters ($\text{\AA}^2 \times 10^3$) for β -PCMOF2 in P3. The anisotropic displacement factor exponent takes the form: $-\pi^2 [h^2 a^{*2} U_{11} + \dots + 2 h k a^* b^* U_{12}]$

Moiety	U11	U22	U33	U23	U13	U12
Na(1)	34(1)	26(1)	24(1)	0(1)	0(1)	17(1)
S(1)	21(1)	18(1)	20(1)	0(1)	0(1)	11(1)
C(1)	11(7)	18(5)	17(3)	0(2)	0(4)	-3(7)
C(2)	15(5)	10(16)	19(3)	3(5)	-3(2)	5(8)
O(1)	21(7)	17(5)	36(3)	-2(2)	-2(4)	13(4)
O(2)	29(3)	25(2)	25(3)	0(2)	-4(2)	16(2)
O(3)	30(3)	25(2)	24(3)	-2(2)	6(2)	14(2)
O(6)	15(3)	20(2)	28(3)	-2(2)	0(2)	6(2)
S(1')	21(1)	16(1)	22(1)	0(1)	0(1)	9(1)
C(1')	7(7)	8(15)	14(3)	-3(5)	-5(4)	4(9)
C(2')	14(6)	19(8)	18(3)	0(4)	1(2)	15(6)
O(1')	20(7)	22(5)	26(3)	0(2)	-4(4)	12(5)
O(2')	27(3)	24(2)	27(3)	5(2)	6(2)	12(2)
O(3')	27(3)	22(2)	25(3)	-7(2)	-6(2)	12(2)
O(6')	14(3)	19(3)	29(3)	-1(2)	0(2)	9(2)

Section S3: PXRD of PCMOFs.

PXRD was used to validate the structures of the as synthesized PCMOFs. In addition, the effects of a) pressing the PCMOF powders into pellets and b) exposing those pellets to impedance analysis were also probed. Data was collected on a Rigaku Multiflex ‘theta-theta’ X-ray diffractometer equipped with a PSC scintillation counter using a $\text{CuK}\alpha$ tube ($\lambda = 1.5418 \text{ \AA}$) operated at 40 kV and 40 mA. Samples for analysis were ground into fine powders using a mortar and pestle. They were then mounted onto a zero background sample holders and the surface of the powders leveled using a glass slide. A scan range of $2\text{-theta} = 3^\circ$ to 60° , with a scan speed of 0.66 deg/min and a sampling width equal to 0.04° were employed. Following collection, the data was processed using winJADE software.

Figure S2: Comparison of pristine β -PCMOF2 and the single crystal of β -PCMOF2. Bottom: PXRD pattern of β -PCMOF2 simulated from the single crystal data. Top: PXRD pattern of pristine β -PCMOF2. Note that the patterns match with the exception of the “hump” at $\sim 20.5^\circ$, which is absent in the simulated pattern. The pristine β -PCMOF2 pattern is representative of that obtained for the hydrated β -PCMOF2 sample.

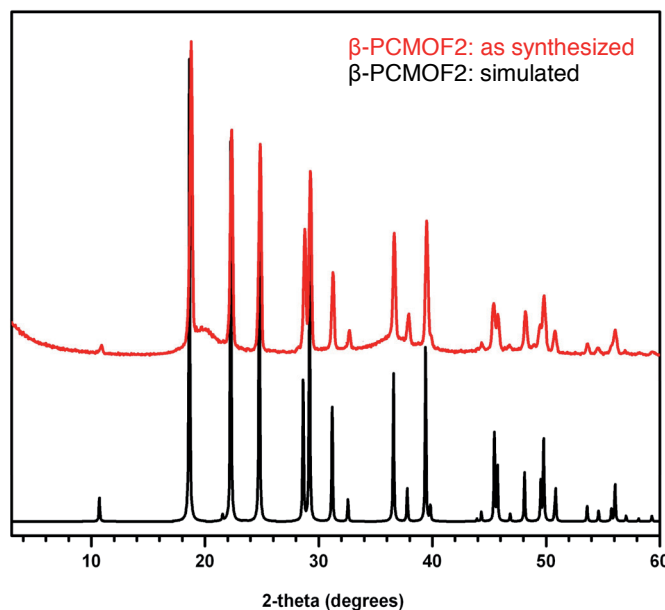


Figure S3: Effect of pellet preparation and impedance analysis on β -PCMOF2. Bottom: PXRD pattern of pristine β -PCMOF2. Middle: PXRD pattern of β -PCMOF2 pressed into a pellet at 5000 kg for 2 minutes. Top: PXRD of the same pellet after impedance analysis. Note that all three patterns match. The patterns are representative of those obtained for the hydrated β -PCMOF2 sample.

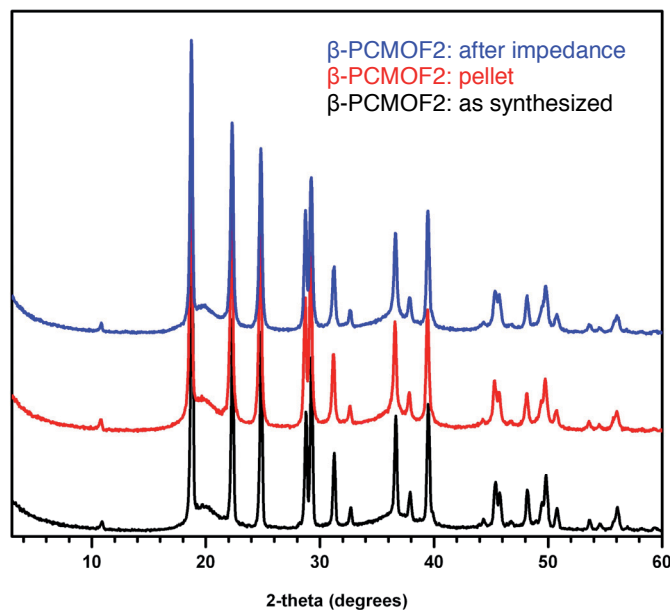


Figure S4: Comparison of the structure of β -PCMOF2(Tz) with β -PCMOF2. Bottom: PXRD pattern of β -PCMOF2 simulated from the single crystal data. Middle: PXRD pattern of pristine β -PCMOF2. Top: PXRD pattern of pristine β -PCMOF2(Tz). Note that all three patterns match.

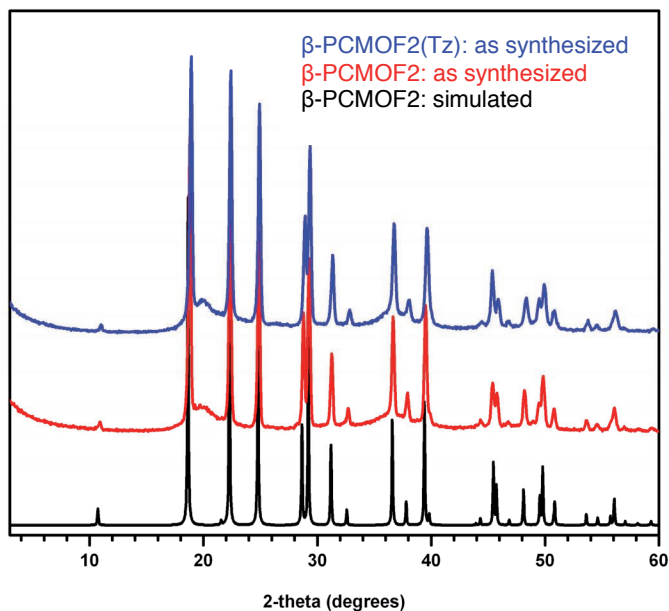


Figure S5: Example of the effect of pellet preparation and impedance analysis on pristine β -PCMOF2(Tz) samples. Bottom: PXRD pattern of pristine $[\beta\text{-PCMOF2(Tz)}_{0.3}]$, **1**. Middle: PXRD pattern of pristine $[\beta\text{-PCMOF2(Tz)}_{0.3}]$, **1** pressed into a pellet at 5000 kg for 2 minutes. Top: PXRD of the same pellet after impedance analysis. Note that all three patterns match. The patterns are representative of those obtained for the $[\beta\text{-PCMOF2(Tz)}_{0.45}]$, **2** and $[\beta\text{-PCMOF2(Tz)}_{0.6}]$, **3** samples.

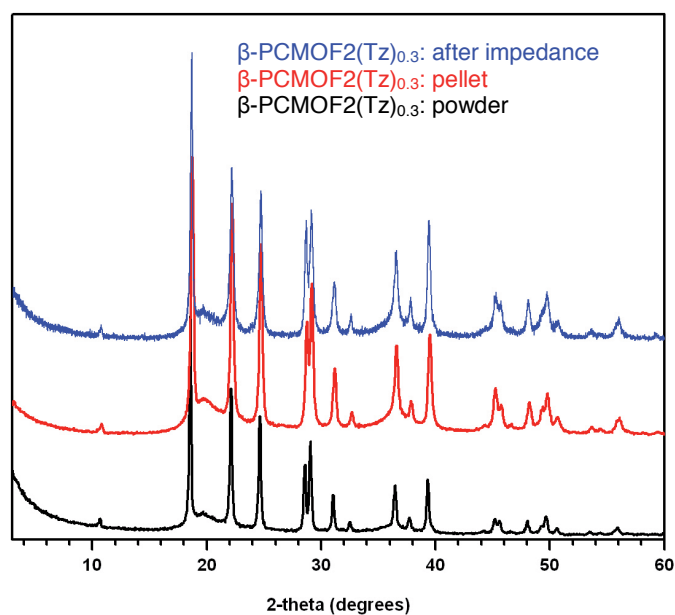
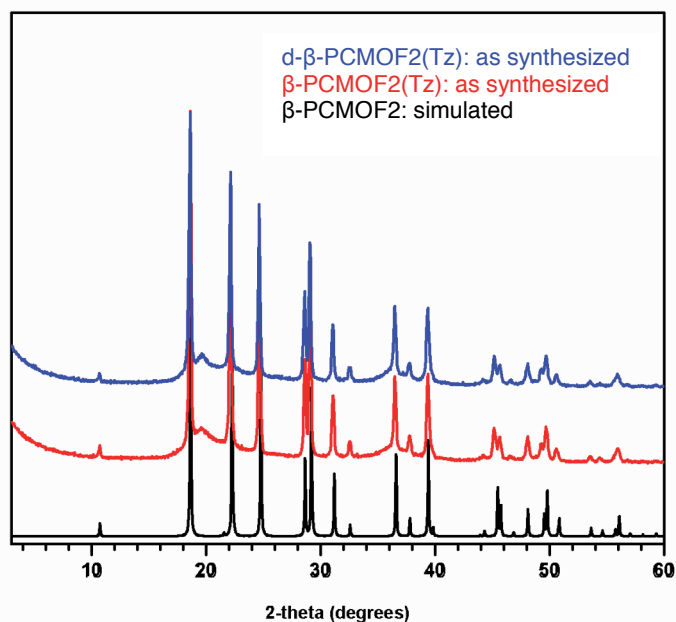


Figure S6: Comparison of the structure of deuterated β -PCMOF2(Tz) [d - β -PCMOF2(Tz)] with β -PCMOF2(Tz). Bottom: PXRD pattern of β -PCMOF2 simulated from the single crystal data. Middle: PXRD pattern of pristine β -PCMOF2(Tz). Top: PXRD pattern of pristine deuterated β -PCMOF2(Tz). Note that all three patterns match.



Section S4: IR, DSC/TGA, Elemental Analysis, ^1H NMR of PCMOFs

Materials and Methods-FT-IR: FT-IR data was used to help verify the identity of the PCMOF materials. The FT-IR spectra were obtained as KBr pellets using a Nicolet Nexus 470 spectrometer.

Figure S7: FT-IR spectrum of pristine β -PCMOF2. The peaks at 3539 and 3430 cm^{-1} corresponds to the presence of water molecules in the framework and aromatic O-H stretching of the ligand.¹ Aromatic C-H stretching is present in the region 2780-2980 cm^{-1} .¹ Bands corresponding to the aromatic ring are observed between 600-800 cm^{-1} and 1300-1630 cm^{-1} .¹ Bands corresponding to the fundamental and split ν_3 S-O stretching modes are found in the region 1000-1240 cm^{-1} .¹

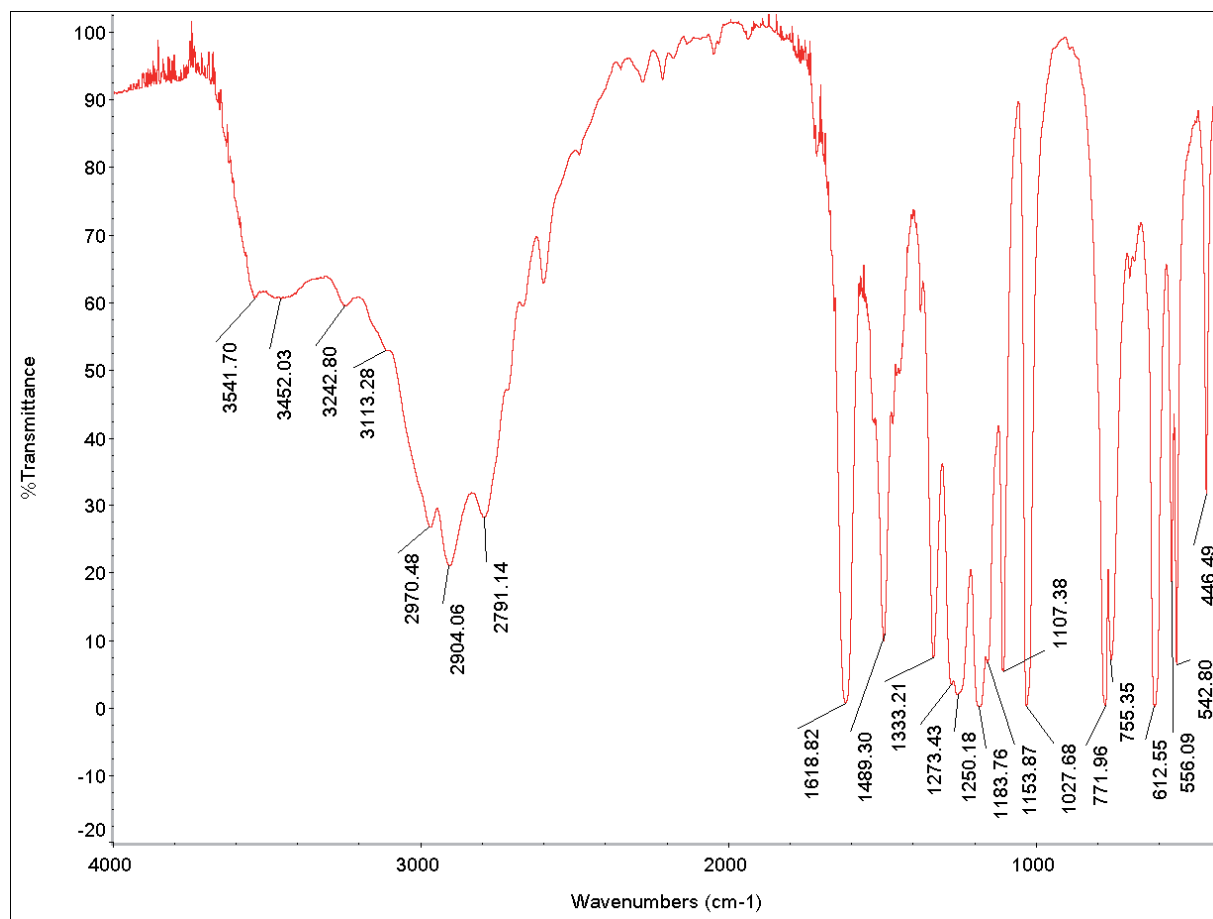
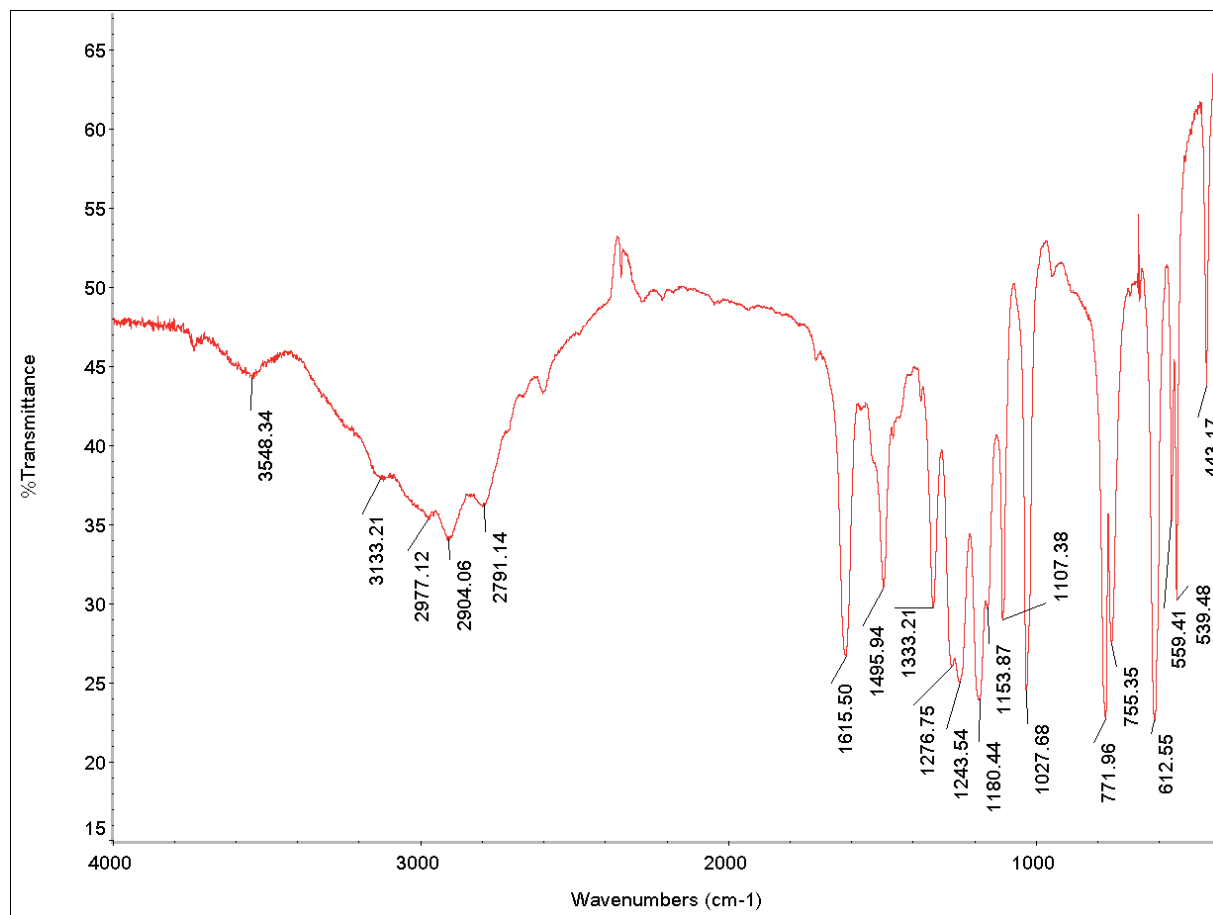


Figure S8: FT-IR spectrum of pristine $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1**. The broad peak at 3548 cm^{-1} corresponds water molecules in the framework, aromatic O-H stretching of the ligand and N-H stretching of 1,2,4-triazole.^{1,2} The band at 3133 cm^{-1} corresponds to C-H stretching of 1,2,4-triazole.² Aromatic C-H stretching is present in the region $2780\text{--}3000\text{ cm}^{-1}$.¹ Bands corresponding to the aromatic ring are observed between $600\text{--}800\text{ cm}^{-1}$ and $1300\text{--}1630\text{ cm}^{-1}$.¹ Finally, bands corresponding to the fundamental and split ν_3 S-O stretching modes are found in the region $1000\text{--}1240\text{ cm}^{-1}$.¹ The spectrum is representative of those obtained for $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2** and $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.



(1): J. Cai, C. Chen, C. Liao, J. Yao, X. Hu and X. Chen, *J. Chem. Soc., Dalton Trans.* **2001**, 1137-1142.

(2): V. A. Lopyrev, N. K. Beresneva, and B. K. Strelets, *Khimiya Geterotsiklicheskikh Soedinenii* **1969**, 5(4), 732-735.

Materials and Methods - Thermal Gravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC), Elemental Analysis (EA) and Proton Nuclear Magnetic Resonance (^1H NMR) of PCMOFs.

DSC/TGA was performed to determine the thermal stability of the PCMOF materials. Crystallographically, the chemical formula of the PCMOF frameworks discussed in this paper (i.e. β -PCMOF2, β -PCMOF2(Tz)_{0.3}, β -PCMOF2(Tz)_{0.45} and β -PCMOF2(Tz)_{0.6}) was determined to be Na_3L , where the ligand (L) is $\text{C}_6\text{H}_3\text{S}_3\text{O}_{12}$. Ea data was also used to confirm this formula, as well as the purity of each of the samples. In addition, EA data in combination with ^1H NMR and TGA data were used to determine the number of guest molecules (i.e. 1,2,4-triazole and/or water) in each framework and thereby derive a formula for each material. TGA/DSC experiments were carried out on a Netzsch 449C Simultaneous Thermal Analyzer in a nitrogen atmosphere. Samples (~ 5 mg of powder) were placed in aluminum pans, and referenced against an empty pan for the DSC measurements. Typical heating programs involved data collection between 25 and 440 °C with heating rates of 5 deg/min. ^1H NMR experiments were performed on a Bruker Avance 400 MHz spectrometer. The 1,2,4-triazole in each sample was quantified by comparing the Tz resonance against that of an acetonitrile internal standard. EA data was obtained on a Control Equipment Corporation Model 440 system, with samples weighed in ambient laboratory atmosphere. All samples were run as duplicates.

Figure S9. Thermal analysis of β -PCMOF2. TGA showed that the material underwent a two-step weight loss. The first step, between 25 and 300 °C is very gradual and resulted from the loss of water from the solid. The second step, beginning at 300 °C, corresponded to thermal degradation of the solid. DSC showed three peaks, broad endothermic peaks at 30 °C and 320 °C and a sharp endothermic peak at 378 °C, corresponding to the weight loss steps in the thermogram. A similar thermogram, with a slightly larger first weight loss step was obtained for the hydrated β -PCMOF2 sample.

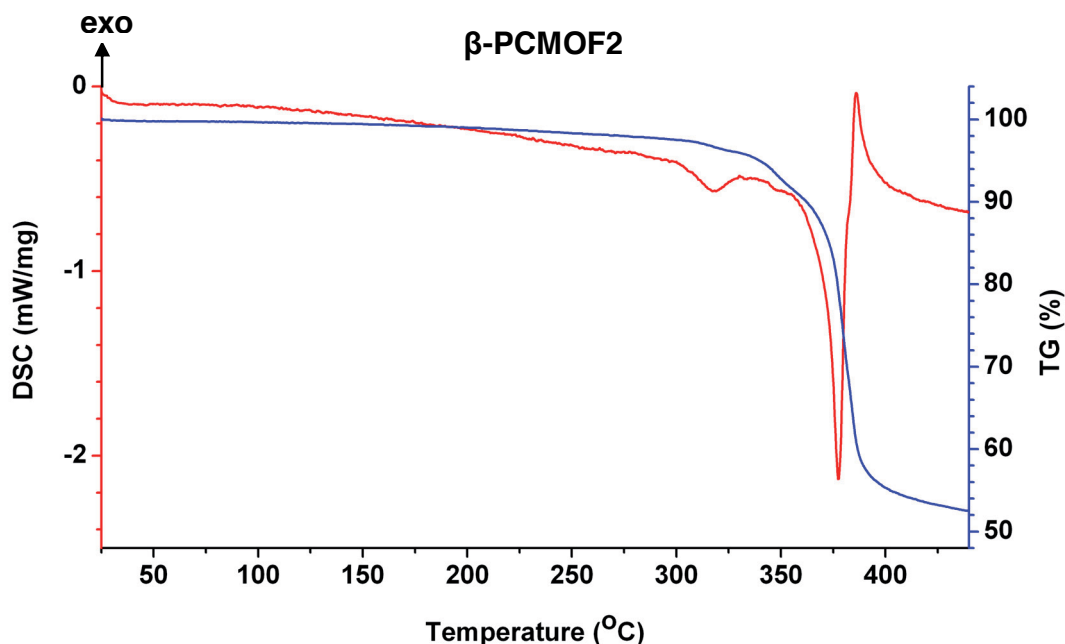


Table S6. Quantifying molecules of guest in β -PCMOF2 with TGA.

Step	Temperature Range (°C)	% weight loss	Number of Guest Molecules
1	25 - 300	2.10	0.5 H ₂ O

Therefore: [Na₃L]•0.5H₂O: Theoretical %C = 16.33, %H = 0.91, %N = 0

Table S7. EA of β -PCMOF2.

Element	Experimental
%C	16.16
%H	0.82
%N	0

The theoretical and experimental values match within reason. Therefore, the formula established for the β -PCMOF2 network was [Na₃L]•0.5H₂O.

*Similarly, the formula established for the hydrated β -PCMOF2 framework was $[\text{Na}_3\text{L}]\cdot 0.6\text{H}_2\text{O}$.

Figure S10. Thermal analysis of $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**. TGA showed a four-step weight loss. The first two weight loss steps from 25 – 100 °C and 100 – 250 °C resulted from the loss of water from the solid. The third weight loss step from 250 – 298 °C, corresponded to loss of 1,2,4-triazole from the framework. The fourth step, beginning at 360 °C, corresponded to thermal degradation of the framework. DSC showed three endothermic peaks, two broad peaks at 40 and 280 °C (corresponding to loss of water and 1,2,4-triazole, respectively), and a sharp endothermic peak at 389 °C (corresponding to framework degradation). Similar thermograms were obtained for both the $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1** and $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3** samples, with variations in mass loss corresponding to the differing amounts of water and 1,2,4-triazole guests in each framework.

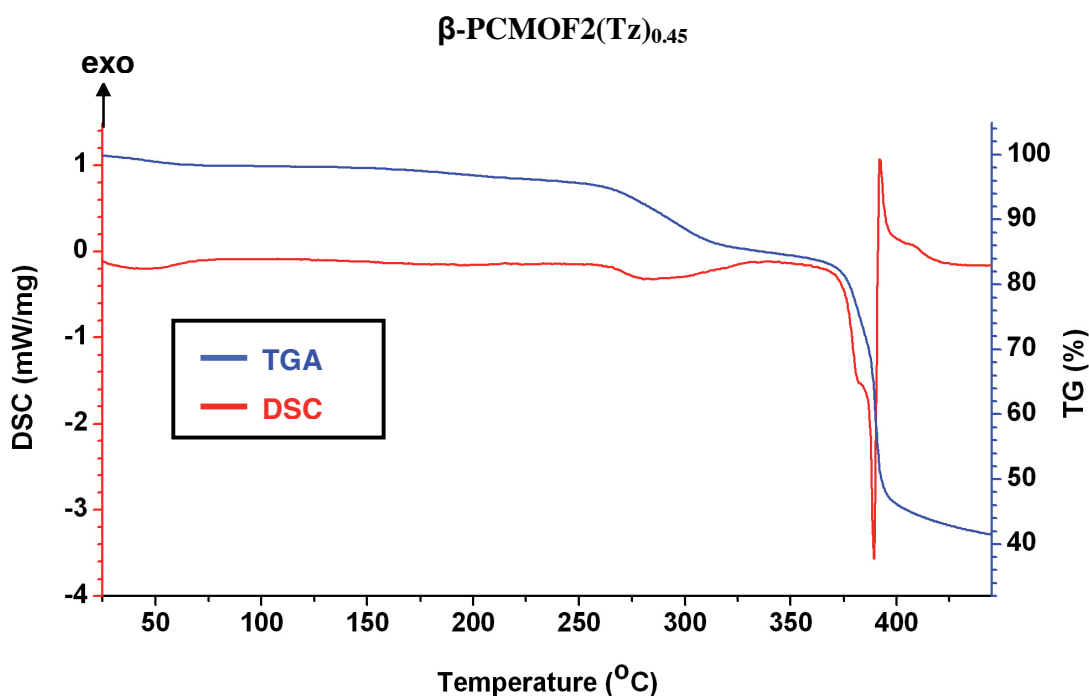


Table S8. Quantifying molecules of guest in β -PCMOF2(Tz)_{0.45} with TGA.

Step	Temperature Range (°C)	% weight loss	Number of Guest Molecules
1	25 - 100	1.66	0.41 H ₂ O
2	100 - 250	2.58	0.64 H ₂ O
3	250 - 298	6.77	0.45 Tz

Therefore: [Na₃L]•0.45(1,2,4-triazole)•1.05H₂O: Theor. %C = 17.19, %H = 1.35, %N = 3.92

Table S9. EA of β -PCMOF2(Tz)_{0.45}.

Element	Experimental
%C	17.25
%H	1.36
%N	3.86

The theoretical and experimental values match within reason. ¹H NMR also validates the ratio of 0.45Tz:ligand. Therefore, the formula established for the β -PCMOF2(Tz)_{0.45} network was [Na₃L]•0.45(1,2,4-triazole)•1.05H₂O.

*Similarly: [β -PCMOF2(Tz)_{0.3}], **1** was found to be [Na₃L]•0.3(1,2,4-triazole)•1.09H₂O.

[β -PCMOF2(Tz)_{0.6}], **3** was found to be [Na₃L]•0.6(1,2,4-triazole)•1.28H₂O.

Section S5: Impedance analysis of PCMOFs.**Materials and Methods- Impedance Analysis of PCMOFs.**

Impedance analysis was performed on pristine PCMOF powders without modification. All pellets were prepared by first grinding the samples into a homogeneous powder with a mortar and pestle. The powders (~0.3 g) were then added to a standard 13 mm die, sandwiched between two porous carbon electrodes (Sigracet, GDL 10 BB, 1.222 cm diameter) and pressed at 5000 kg for 2 min.

β -PCMOF2 was also analyzed by first hydrating the solid prior to preparation of the pellet. In this procedure, approximately 0.3 g of sample was placed into a 30 mL beaker, the beaker placed into an 8 oz jar with 60 mL distilled water and the jar sealed for 12 h. Pellets of the hydrated powder was then prepared as outlined above.

All pellets were 13 mm in diameter and ranged in thickness from 1.30 to 1.38 mm. Measurements were performed using an impedance and gain-phase analyzer (SI model 1260), 1 Hz - 1 MHz, with a two-probe electrochemical cell equipped with platinum current collectors and an applied AC voltage of 10 - 100 mV (unless otherwise stated). Measurements were taken in the temperature range of 23 to 150 °C in an anhydrous H₂ or D₂ atmosphere. Conductivity was calculated using the following equation : $\sigma = l/R_s S$, where l and S are the thickness (cm) and cross-sectional area (cm²) of the pellet respectively, and R_s is the bulk resistivity of the sample (Ω), which R_s was extracted directly from the impedance plots.

Figure S11. Comparison of the Arrhenius plots for β -PCMOF2 measured in an anhydrous H_2 atmosphere with and without initial sample hydration. The procedure for hydrating the sample was outlined in the *Materials and Methods, Section S5*. Red circles = powder was hydrated prior to preparation of the pellet for analysis; blue triangles = as synthesized material. The higher conductivity of the hydrated sample, suggests that water functions as a proton carrier in this system. As expected, with heating, water is lost from the framework and conductivity decreases.

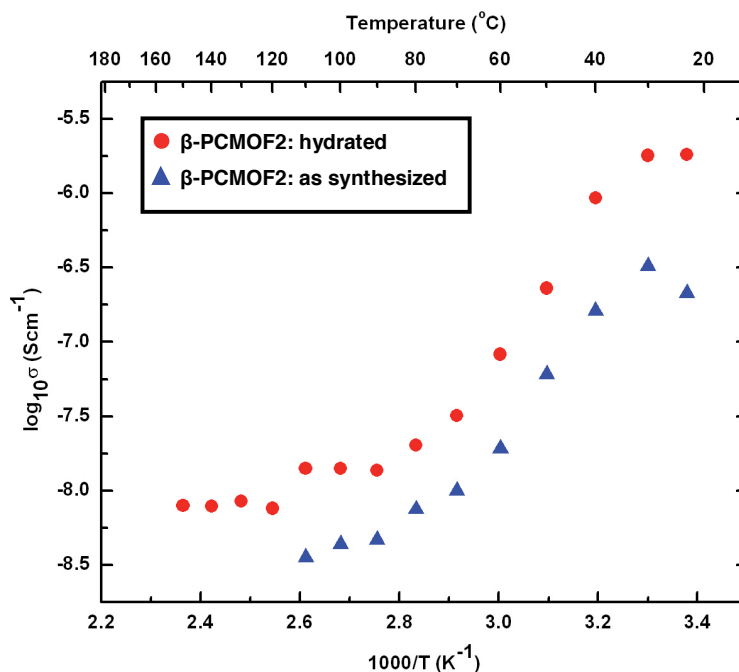
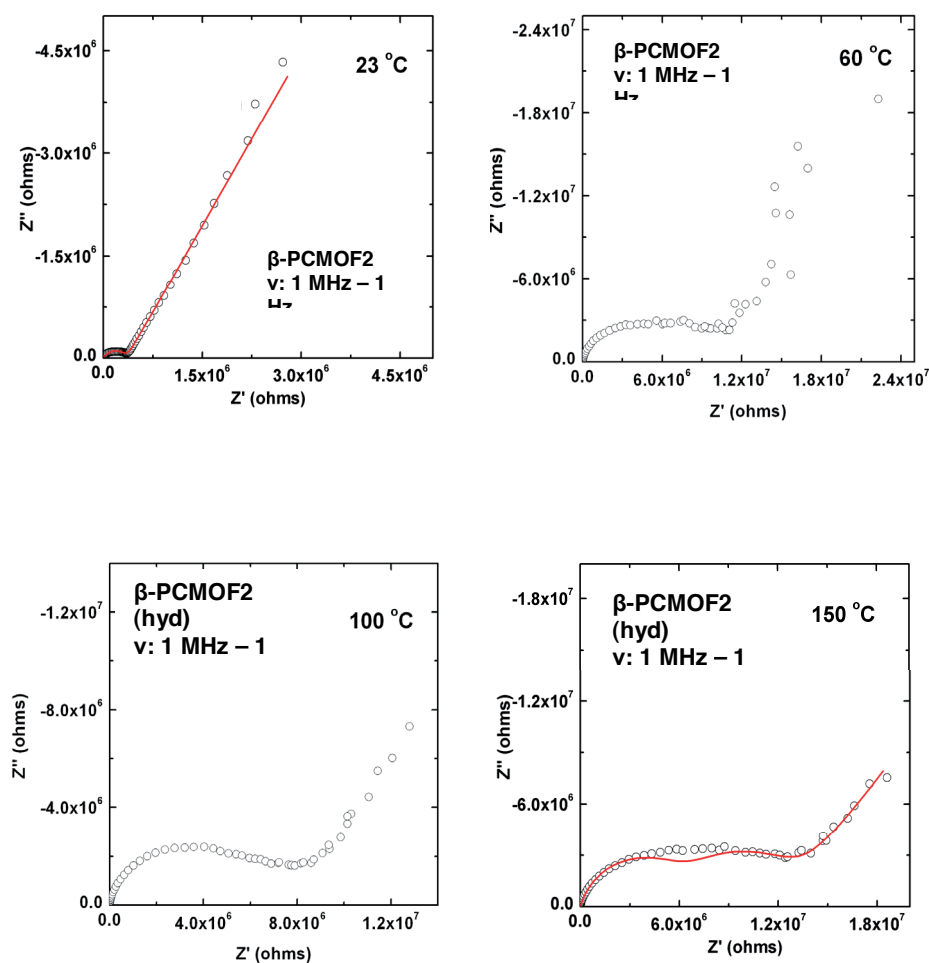
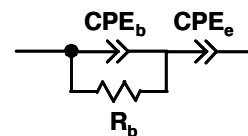


Figure S12. Representative Nyquist plots of β -PCMOF2 measured in anhydrous H_2 with and without initial sample hydration. The procedure for hydration is given in *Section S5*. The lines through the points of the 23 °C and 150 °C plots represent fits for the following equivalent circuits: 23 °C, $(R_bCPE_b)(CPE_{cl})$; 150 °C, $(R_bCPE_b)(R_{gb}CPE_{gb})(CPE_{cl})$.



Sample	T (°C)	R_b	CPE_b-T	CPE_b-P	CPE_e-T	CPE_e-P
β -PCMOF2	23	351560	1.2×10^{-9}	0.65	6.2×10^{-8}	0.66



Sample	T (°C)	R_b CPE_b-T CPE_b-P	R_{gb} $CPE_{gb}-T$ $CPE_{gb}-P$	CPE_e-T CPE_e-P
β -PCMOF2-hyd	150	6.22×10^6 5.0×10^{-11} 0.83	6.18×10^6 1.0×10^{-9} 0.81	3.5×10^{-8} 0.58

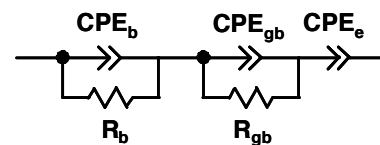


Figure S13. Comparison of the Arrhenius plots for $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1**, $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**, $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**, measured in anhydrous H_2 atmospheres. $\beta\text{-PCMOF2}(\text{Tz})_{0.3}$: heating cycle 1 = solid green circles, cooling cycle 1 = open green circles, heating cycle 2 = solid green triangles, cooling cycle 2 = open green triangles. $\beta\text{-PCMOF2}(\text{Tz})_{0.45}$: heating cycle 1 = solid blue circles, cooling cycle 1 = open blue circles, heating cycle 2 = solid blue triangles, cooling cycle 2 = open blue triangles. $\beta\text{-PCMOF2}(\text{Tz})_{0.6}$: heating cycle 1 = solid red circles, cooling cycle 1 = open red circles, heating cycle 2 = solid red triangles, cooling cycle 2 = open red triangles. Note for all 3 samples: after each heating cycle, the sample temperature was maintained at 150 °C for 12 – 18 hours, prior to taking measurements for the subsequent cooling cycle. Also the sample temperature was maintained at 23 °C for ~12 h after cooling cycle 1, prior to taking measurements for heating cycle 2.

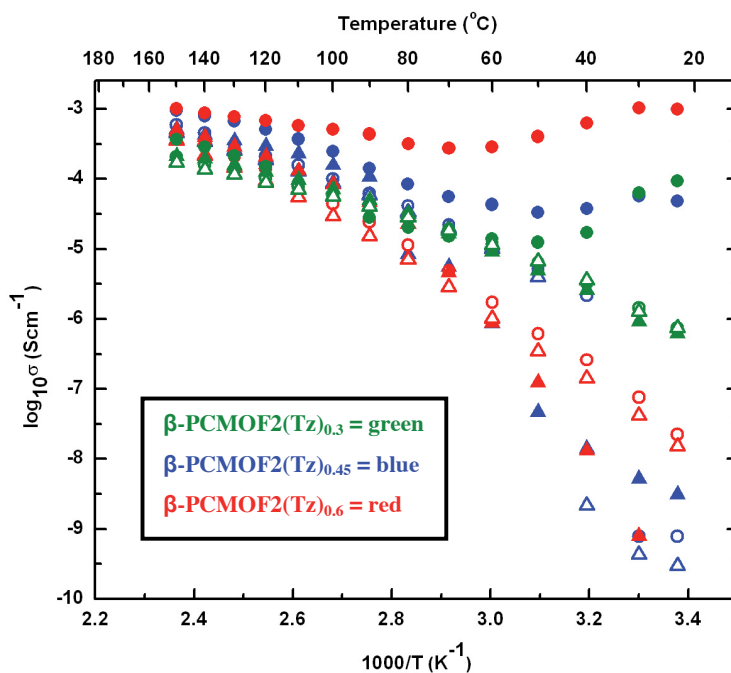
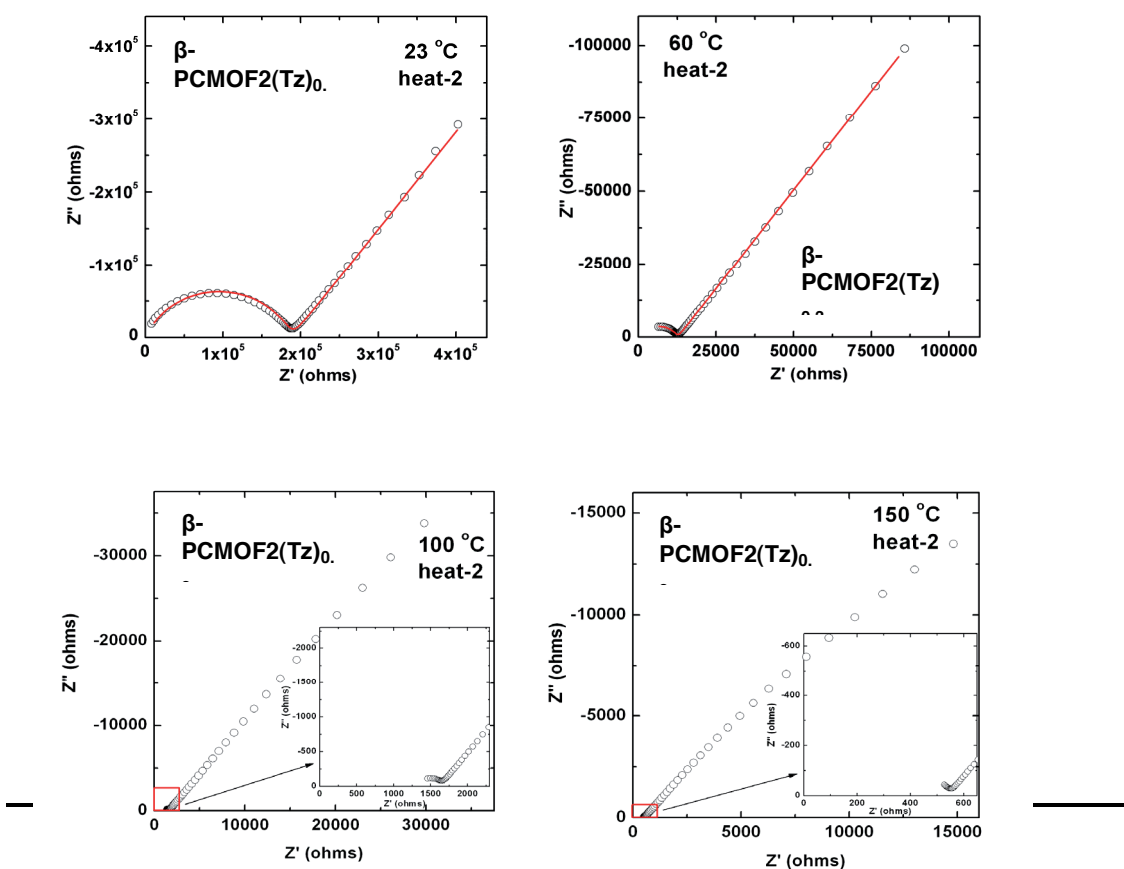
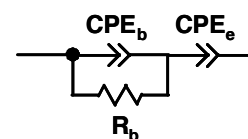


Figure S14. Sample Nyquist plots of the second heating cycle for $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$,

1. These plots are representative of the heating and cooling cycles of all three $\beta\text{-PCMOF2}(\text{Tz})_x$ samples (where $x = 0.3, 0.45, 0.6$). Measurements were performed in anhydrous H_2 . The lines passing through the data points of the plots at 23 °C and 60 °C represent fitted lines for the equivalent circuit $(R_b\text{CPE}_b)(\text{CPE}_{e1})$.

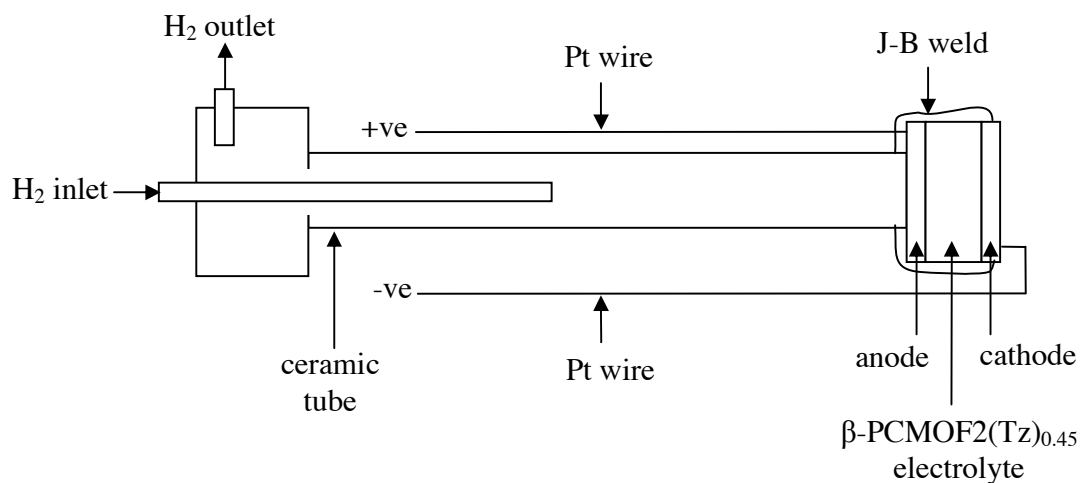


Sample	T (°C)	R_b	$\text{CPE}_b\text{-T}$	$\text{CPE}_b\text{-P}$	$\text{CPE}_e\text{-T}$	$\text{CPE}_e\text{-P}$
$\beta\text{-PCMOF2}(\text{Tz})_{0.3}$	23	188070	3.3×10^{-10}	0.75	9.5×10^{-7}	0.59
$\beta\text{-PCMOF2}(\text{Tz})_{0.3}$	60	12701	2.4×10^{-9}	0.66	2.8×10^{-6}	0.595



Section S6: EMF Study of PCMOFs.**Materials and Methods – Electromotive Force Studies of $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**.**

Electromotive force studies were carried out on $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**, as proof of principle of the ability for these materials to conduct protons. A membrane electrode assembly (MEA) was first prepared following a slightly modified procedure to that used in the preparation of the samples for impedance analysis. First, pieces of platinum wire (~15–20 mm in length) were tied to porous carbon electrodes (Sigracet, GDL 10 BB, 1.3 cm diameter) and coiled around the perimeter of each electrode cut out. Ground $\beta\text{-PCMOF2}(\text{Tz})_{0.45}$ powder (0.4015 g) was then sandwiched between the two electrodes in a 13mm die (with the coiled pieces of platinum wire facing away from the sample) and pressed at 5000 kg for 2 minutes. The pieces of platinum wire were then lifted off the two sides of the MEA and functioned as electrical contacts to the anode and cathode during the measurements. A hollow piece of ceramic tube was then sealed onto one side of the MEA using a two component epoxy (J-B weld). The other end of the ceramic tube was attached to a gas inlet-outlet apparatus and sealed with J-B weld. This allowed H_2 gas to be passed through the ceramic tube and selectively over the sealed side of the assembly, which functioned as the anode. The other side of the MEA was left open to the air and functioned as the cathode. Open circuit voltage (OCV) measurements were performed with a Solartron SI 1287 electrochemical interface. CorrWare and CView2 software were used to collect and analyze the data.

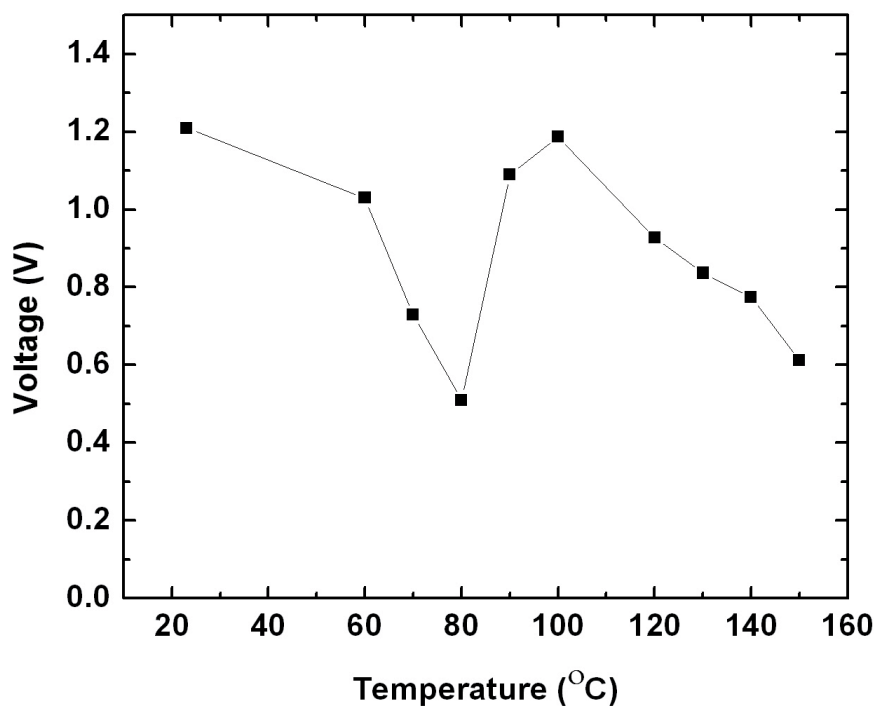
Figure S15. Schematic of the device used for EMF measurements.**Table S10.** Open circuit voltage of [β -PCMOF2(Tz)_{0.45}], **2**.

Temperature (°C)	Voltage (V)	Duration of Time Held at Temperature (h)
23	1.21	12
60	1.03	13
70	0.73	70
80	0.51	2.5
90	1.09	40*
100	1.18	72
120	0.93	18.5*
130	0.84	5*
140	0.77	4.5
150	0.61	2.75**

* non-equilibrated voltage: slowly increasing

** non-equilibrated voltage: slowly decreasing (due to fuel crossover)

Figure S16. Open circuit voltage of $[\beta\text{-PCMOF2(Tz)}_{0.45}]$, **2**. The initial loss in voltage observed from 23 °C to 80 °C is believed to arise from dehydration of the sample. The sharp increase in voltage at 90 °C is thought to be due to the contribution of the 1,2,4-triazole guests in the framework, which are more kinetically active at this temperature. A similar phenomenon is observed in the ac impedance studies (*Section S5, Figure S12*), where the second heating cycle of the $\beta\text{-PCMOF2(Tz)}_{0.45}$ sample, showed a marked increase in conductivity at 90 °C.



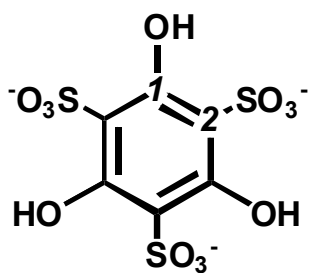
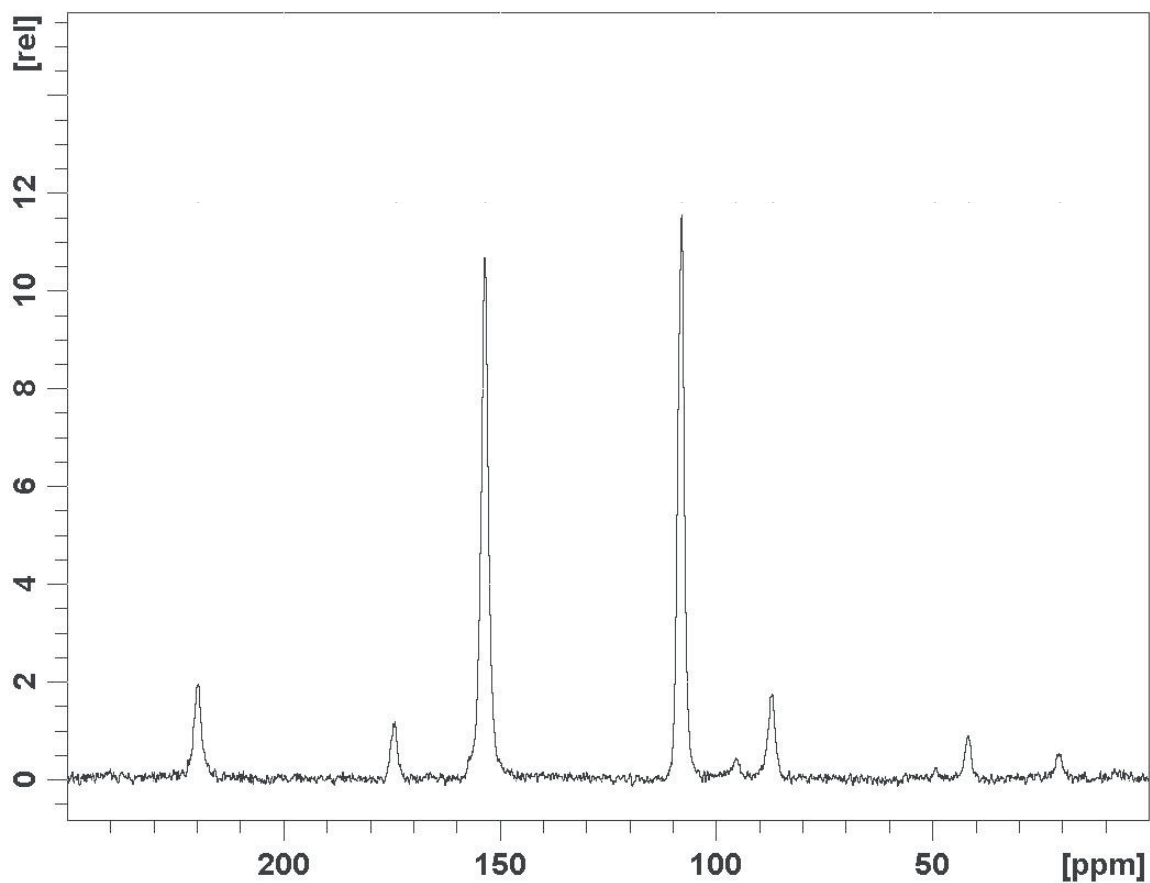
Section S7: SS NMR of PCMOFs.

Materials and Methods - Solid-State ^{13}C CP/MAS Nuclear Magnetic Resonance Studies (NMR) of β -PCMOF2, $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1**, $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**, $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.

Solid-state ^{13}C CP/MAS NMR spectra were collected at ambient temperature using a Bruker AMX-300 spectrometer with a Bruker magic angle spinning (MAS) BL-4 probe and 4 mm zirconia rotors at 75.47 MHz. Samples were spun at the magic angle at a rate of 5 kHz and required collection times ranging between 2 and 6 h. The ^{13}C chemical shifts were externally referenced to the carbonyl signal of lysine ($\text{C}=\text{O}$ δ_{iso} 176.3 ppm).

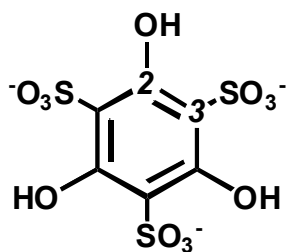
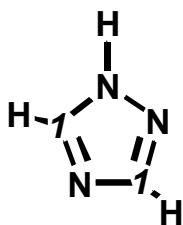
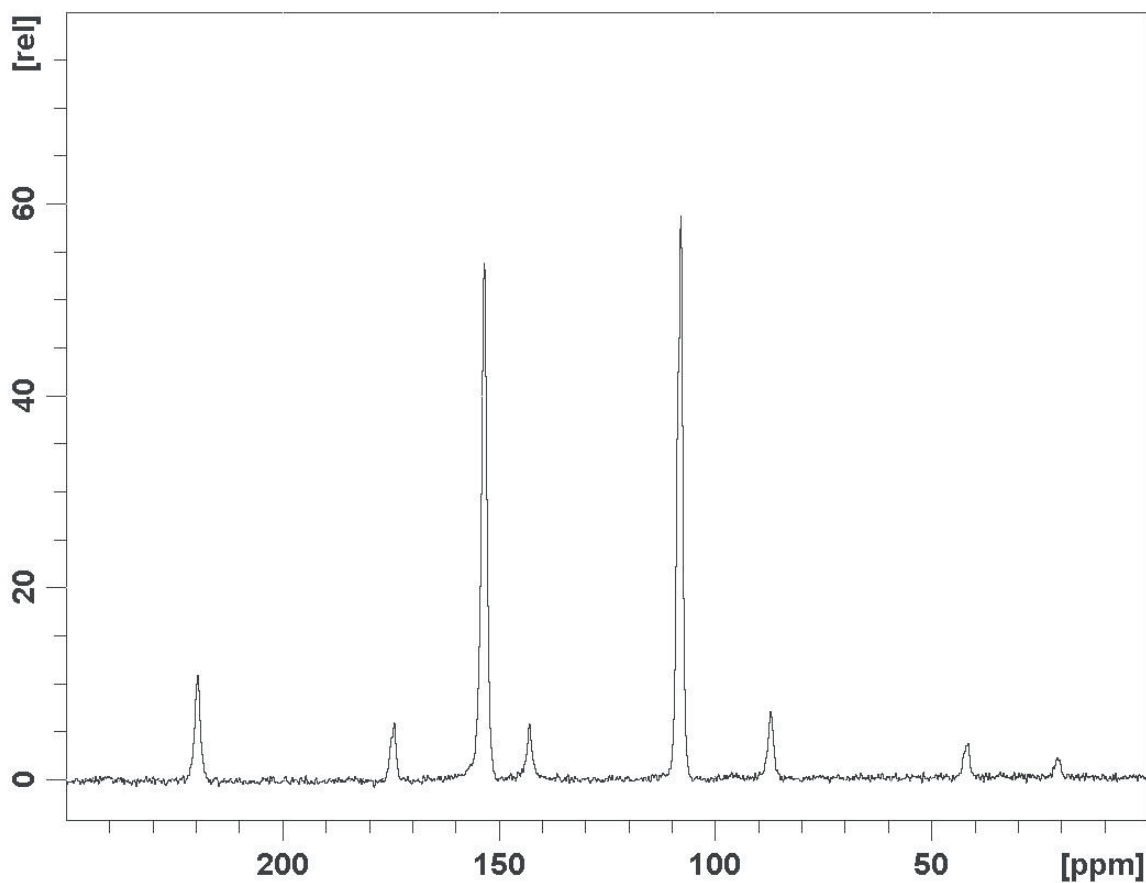
Solid state ^1H MAS NMR spectra were obtained at various temperatures using a Bruker DSX 400 spectrometer with a Bruker MAS BL4 probe and 4mm thick wall zirconia rotors (Spin rate ~ 16 kHz, reference TMS). Temperatures were corrected for frictional heating using a calibration of the ^{207}Pb resonance in $\text{Pb}(\text{NO}_3)_2$ [A. Bielecki, A.; Burum, D. *J. Magn. Res.* **1995**, *116*, 215].

Figure S17: CPMAS Solid-State ^{13}C NMR spectrum of β -PCMOF2 and peak assignment. Unassigned peaks correspond to spinning side bands.



Carbon	Chemical Shift in ppm
1	107.91
2	153.42

Figure S18: CPMAS Solid-State ^{13}C NMR spectrum of $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2** and peak assignments. Unassigned peaks correspond to spinning side bands. The spectrum is representative of those obtained for $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1** and $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.



Carbon	Chemical Shift in ppm
1	142.90
2	107.85
3	153.32

Figure S19. Solid state ^1H MAS NMR spectra of (from top to bottom) beta-PCMOF2(Tz)_{0.6}, beta-PCMOF2(Tz)_{0.3}, bulk 1,2,4-triazole and beta-PCMOF2, (spinning at 16kHz the temperature is 318.1 K). The signal at 11.8 ppm can be assigned to the O-H on the PCMOF2 aromatic ring. The signals at ~9.2 ppm arise from the C-H of incorporated Tz. The weaker signals near 4 ppm and 5 ppm are most probably water molecules.

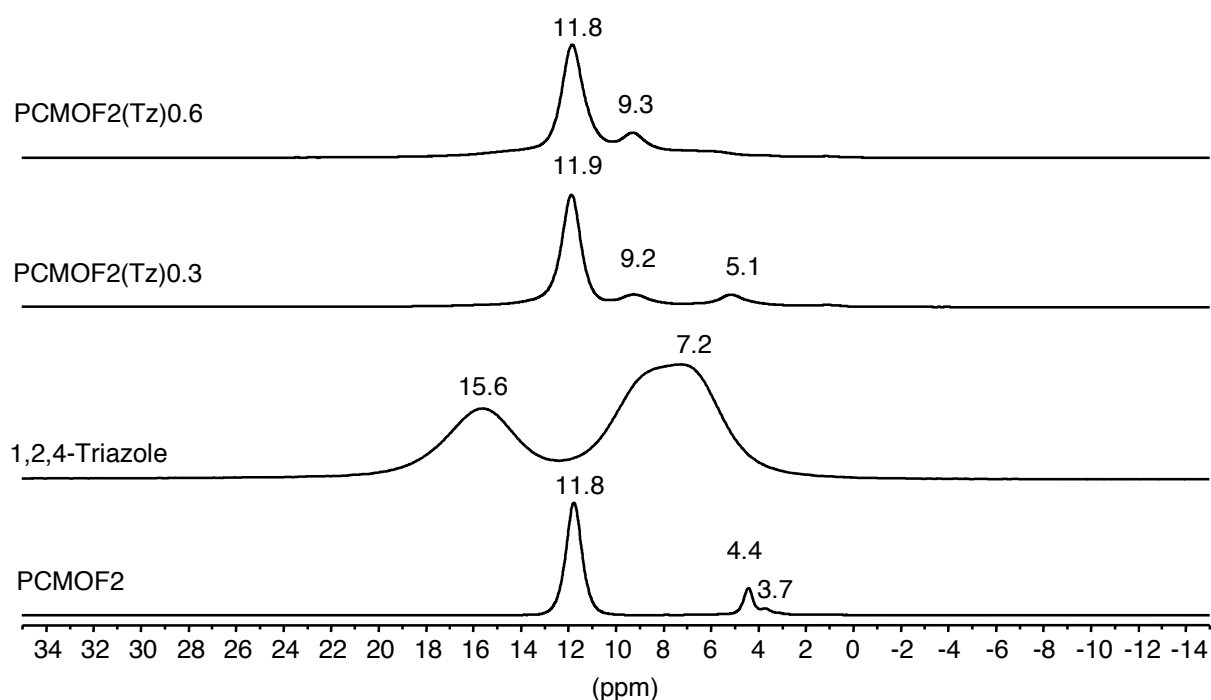


Figure S20. Solid state ^1H MAS NMR spectra of bulk 1,2,4-Triazole as a function of temperature (MAS ~ 16 kHz). At 318K the spectrum consists of a singlet at 15.6 ppm which can be assigned to N1-H and an unresolved doublet centred at 7.9 ppm which can be assigned to C3-H (low field) and C5-H (high field). The presence of the two C-H resonances is consistent with the asymmetric tautomer shown in the crystal structure (Goldstein et al.). The larger shift for N1-H in the solid reflects the hydrogen bonding to N4 on an adjacent molecule observed in the crystal structure. The two sharp lines in the 392K spectrum represent the liquid phase produced by overheating before temperature equilibration, and the shifts correspond with values observed previously (Creagh et al.). The single resonance for C-H indicates tautomerism.

Goldstein, P., Ladell, J., Abowitz, G. Refinement of the Crystal and Molecular Structure of 1,2,4-Triazole ($\text{C}_2\text{H}_3\text{N}_3$) at Low Temperature. *Acta Cryst.* **B25**, 135-143 (1969).

Creagh, L. T., Truitt, P. Nuclear Magnetic Resonance Studies of Triazoles. I.

Tautomerism of 1,2,4-Triazole. *J.Org.Chem.* **33**, 2956-2957 (1968).

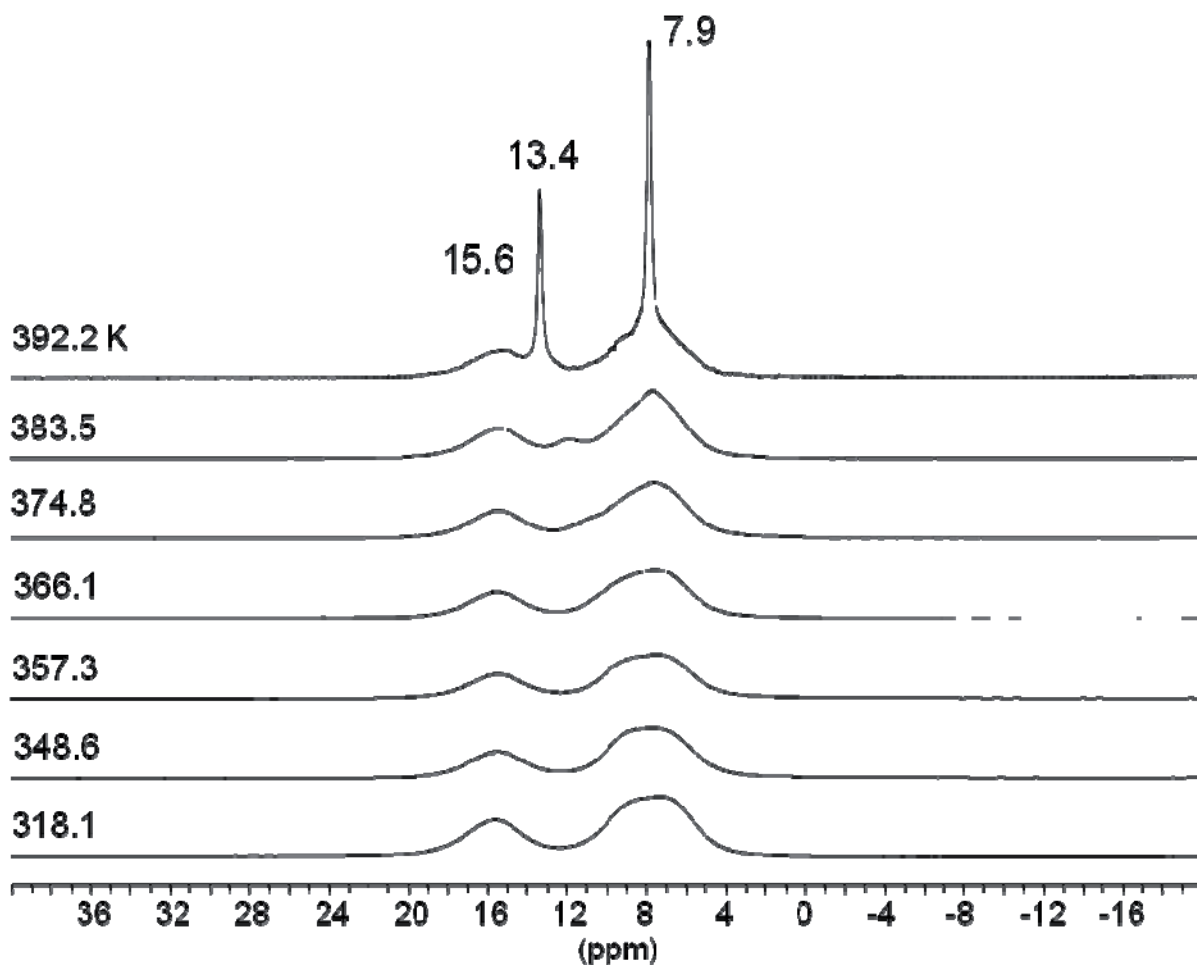


Figure S21. Solid state ^1H MAS NMR spectra of $\beta\text{-PCMOF2}(\text{Tz})_{0.3}$ as a function of temperature (MAS ~ 16 kHz). This material does not appear to show any resonances from bulk Tz. The line at 9.1 ppm is best assigned to the C-H of the triazolium cation (shifted markedly from 7.9 ppm observed for triazole) and the single line indicates tautomerism. The lines sharpen with increasing temperature and the asymmetric 11.7 ppm line reveals a shoulder and then a peak at 10.9 ppm which presumably is due to N-H of triazolium. The resonances of triazolium cation in CD_3CN solution have been reported at 9.89 ppm (C-H) and 10.47 ppm (N-H) (Moers et al.).

Moers, O., Wijaya, K., Jones, P. G., Blaschette, A. Polysulfonylamines. CIX. 1,2,4-Triazolium di(methanesulfonyl)amidate. *Acta Cryst.* **C55**, 754-756 (1999).

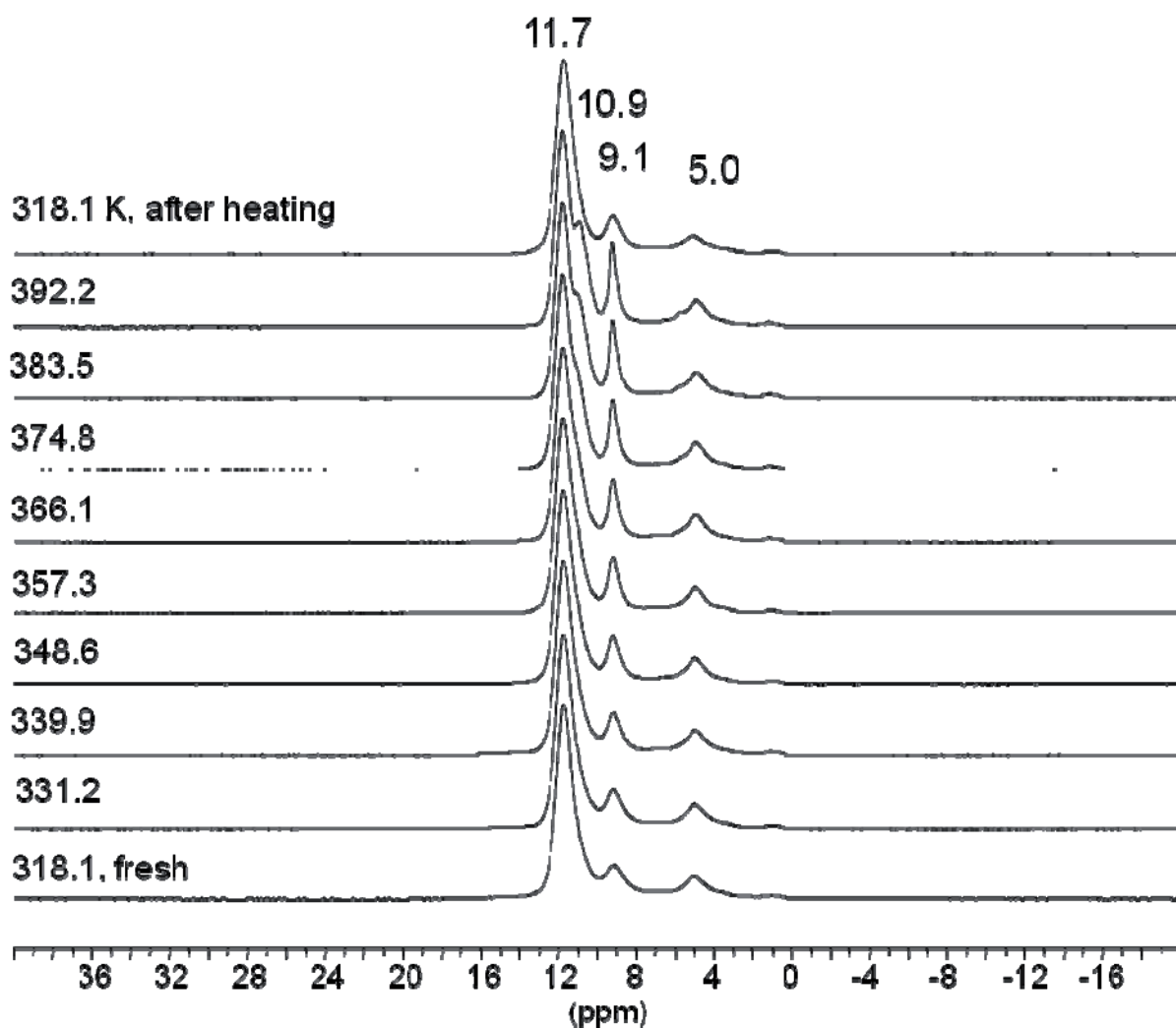
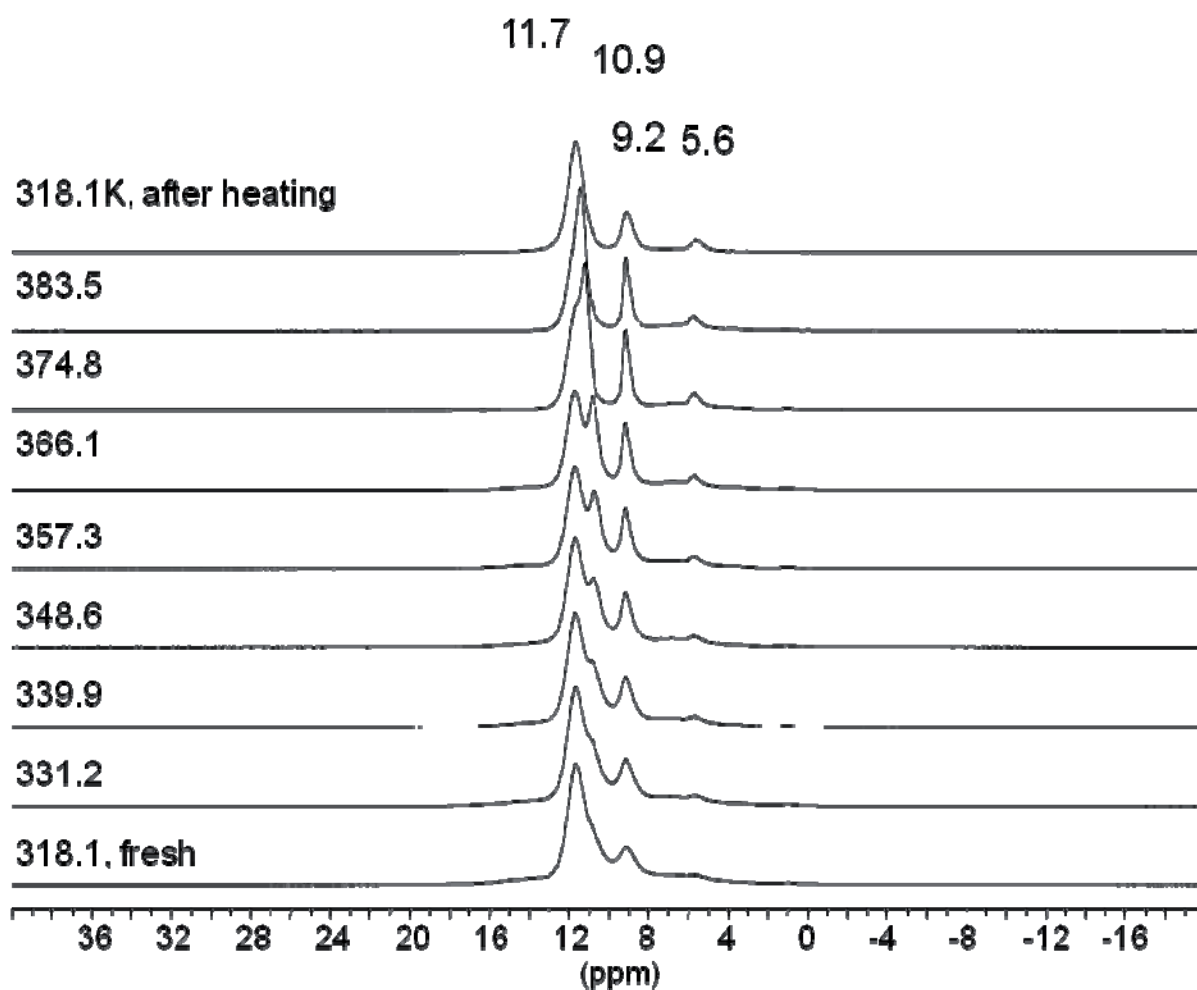


Figure S22. Solid state ^1H MAS NMR spectra of $\beta\text{-PCMOF2}(\text{Tz})_{0.6}$ as a function of temperature (MAS ~ 16 kHz). The fresh sample shows traces of the broad signals of bulk Tz underlying the sharper signals of the PCMOF2 and the Tz inside the pores. These bulk signals disappear on heating, the Tz presumably absorbs into the PCMOF2 and stays inside on cooling. The increased intensity of the 9.2 ppm C-H line relative to the 11.7

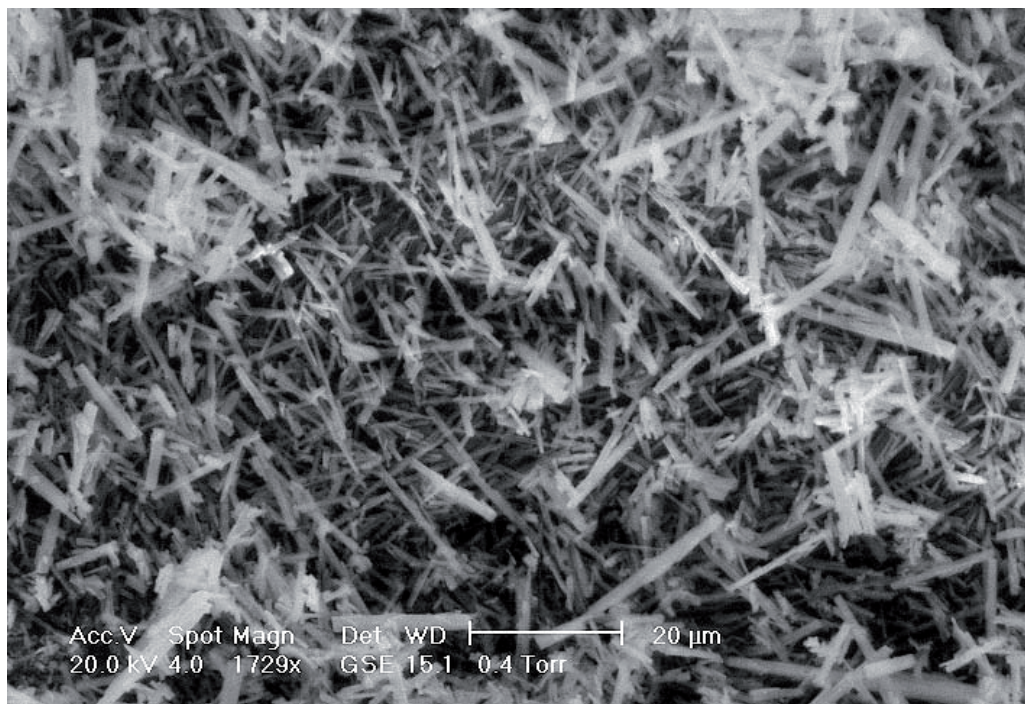


Section S8: SEM of PCMOFs.***Materials and Methods - Scanning Electron Microscopy Imaging (SEM) of PCMOFs.***

SEM was used to determine the morphology of the crystals in each of the PCMOF products. Samples of the PCMOFs were analyzed and their purity verified with the presence of a single crystal morphology for each material. Samples for analysis were prepared by attaching a piece of sticky carbon onto flat aluminum sample holders and then coating these surfaces with the PCMOF materials. Samples requiring gold coating were prepared using a Hummer sputter coater at 50 millitorr of pressure in an argon atmosphere for 3 minutes, with 15 mA of current. Samples were analyzed on a Philips XL30 ESEM, utilizing either a GSE detector with water vapor pressures between 0.4 and 1.0 Torr and an accelerating voltage of 20 KV (samples run using the environmental mode of the instrument) or an SEI detector and an accelerating voltage of 20 KV (gold coated samples). The environmental mode was utilized, except for the 1,2,4-triazole loaded samples after impedance, to eliminate the need for gold coating the samples, thereby preventing charging of the materials during analysis.

Figure S23: SEM images of β -PCMOF2 at different magnifications illustrating a rod-shape morphology. Note the homogeneity/phase purity of the sample.

(a)



(b)

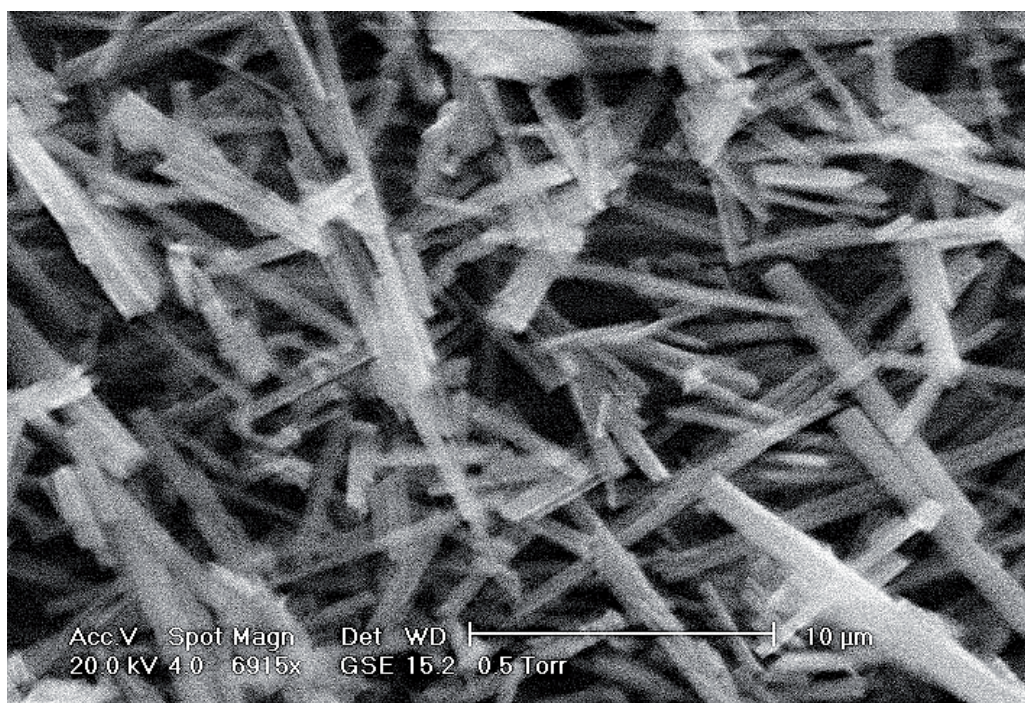
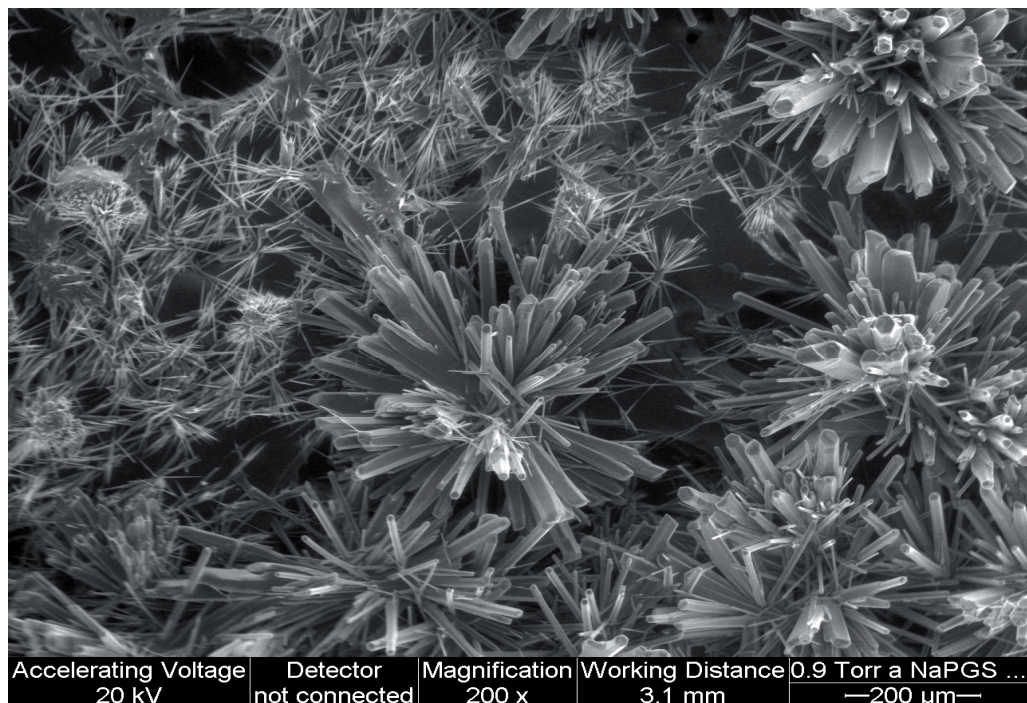


Figure S24: SEM images of $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2** at different magnifications illustrating a rod-shape morphology. Note the homogeneity/phase purity of the sample. The micrographs are representative of those obtained for $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1** and $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.

(a)



(b)

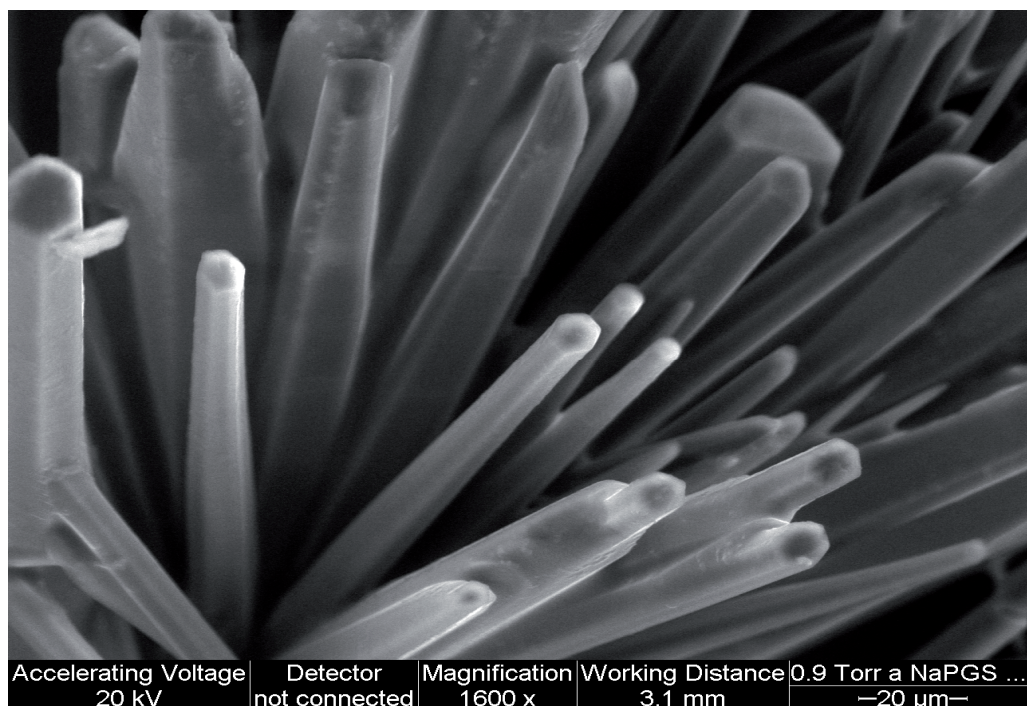


Figure S25: SEM image of $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2**, following impedance analysis. It is evident that the crystals still possess a rod-shaped morphology but that they are much smaller due to grinding of the powder during pellet preparation. Gold coated sample. The micrograph is representative of those obtained for $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1** and $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.

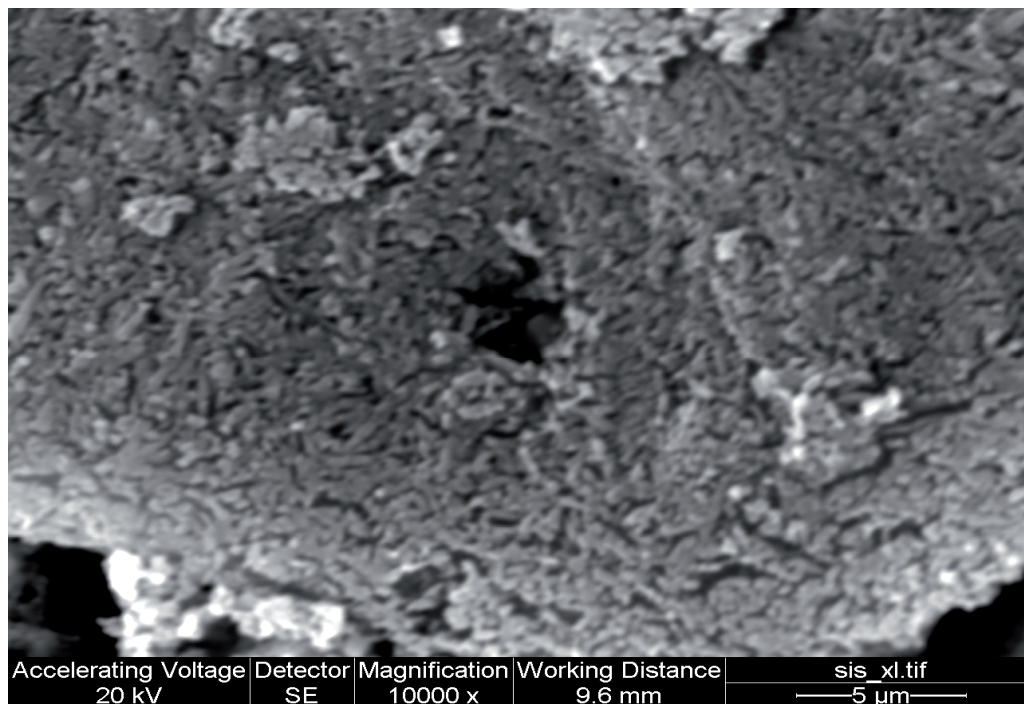
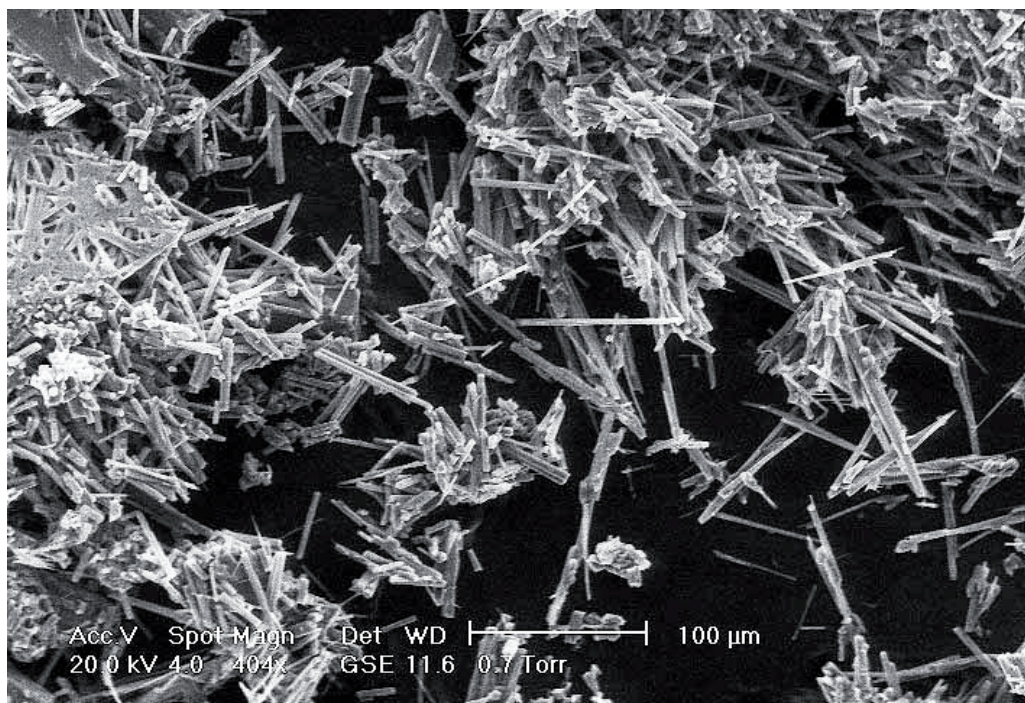


Figure S26: SEM images of recrystallized $[\beta\text{-PCMOF2}(\text{Tz})_{0.3}]$, **1**, at different magnifications following impedance analysis. It is evident that the rod-shaped morphology of the sample prior to impedance analysis has been reestablished. The micrographs are representative of those obtained for $[\beta\text{-PCMOF2}(\text{Tz})_{0.45}]$, **2** and $[\beta\text{-PCMOF2}(\text{Tz})_{0.6}]$, **3**.

(a)



(b)

